# On the Possibility of Informationally Inefficient Markets Without Noise

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**Abstract**

Full revelation of information through prices in rational-expectations models is a knife-edge property of exponential (CARA) utility. I show that partial revelation arises under any other expected-utility preference—including CRRA—without noise traders and without any substitute: no random endowments, no supply shocks, no hedging motives, no behavioural biases. The model has zero exogenous noise of any kind. The mechanism is an aggregation-space mismatch: CARA clears markets in log-odds space, which coincides with the Bayesian sufficient statistic, so prices reveal the statistic exactly. Non-CARA preferences aggregate in a different space, and the resulting Jensen gap breaks full revelation. I prove this for the no-learning equilibrium, establish that CARA is the unique preference class with log-odds-linear demand, characterise the rational-expectations equilibrium via a contour-integration fixed point, and verify that partial revelation survives learning from prices. The Grossman–Stiglitz paradox is resolved within the rational class.

**Keywords:** rational expectations, partial revelation, noise traders, CRRA, Grossman–Stiglitz paradox, aggregation space.
**JEL codes:** D82, D83, G14.

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# 1 Introduction

Half a century after Grossman and Stiglitz (1980), the textbook resolution of the informational paradox in asset markets remains the same: prices cannot aggregate private signals perfectly because noise — exogenous shocks to supply, liquidity demand, or trader population — masks the information content of order flow. From Hellwig (1980) and Diamond and Verrecchia (1981) through Kyle (1985) and the modern microstructure literature, the conditional partial revelation of information is engineered by an unmodelled population whose trades are unrelated to fundamentals. The modelling convention is so entrenched that introductory treatments routinely list "noise traders" alongside risk aversion and information asymmetry as a *primitive* of the rational-expectations framework.

This paper makes a simple but consequential point: that primitive is not necessary. In a fully rational economy with no liquidity shocks, no behavioural traders, and zero net supply, partial revelation arises endogenously from preferences alone. What is required is that agents do not have constant absolute risk aversion. Once preferences exit the CARA class, the mapping from private information to equilibrium prices fails to identify the Bayesian sufficient statistic, and prices reveal partial information for purely aggregational reasons.

The mechanism is best stated in terms of an *aggregation space*. Under any preference class, equilibrium prices aggregate private posteriors via market clearing. Under CARA, demand is linear in the log-odds of the posterior, and clearing therefore averages log-odds. The Bayesian sufficient statistic in the canonical Gaussian-signal model is itself a sum of log-likelihood ratios. The two spaces coincide — log-odds in, log-odds out — and prices fully reveal that statistic. Under any other CRRA preference, demand is nonlinear in log-odds and approximately linear in probability; clearing therefore averages in probability space. The probability average of a logistic transform is not the logistic transform of the average,

$$\frac{1}{K}\sum_{k=1}^{K}\Lambda(\tau u_k) \neq \Lambda\left(\frac{1}{K}\sum_{k=1}^{K}\tau u_k\right),$$

and the Jensen gap on the right is the size of the revelation failure. CARA is the boundary case in which the gap closes because the link from posteriors to demand is itself the inverse logistic — a coincidence of functional form, not an economic primitive.

The paper makes one conceptual and four formal contributions. The conceptual contribution is to recast the question of revelation in preference-theoretic rather than friction-theoretic terms: in any rational-expectations economy without noise, full revelation requires that the demand functional aggregate private posteriors linearly in log-odds, and CARA is the unique CRRA preference class for which this is the case. We refer to this requirement as the *alignment principle* in Section 2.4. Once stated cleanly, the noise-trader assumption appears as a device

that perturbs CARA enough to break the alignment artificially; non-CARA preferences break the same alignment endogenously and at no modelling cost.

Building on this principle, the paper makes four formal contributions. *First*, in a binary-state model with $K \geq 3$ informed groups, I show that the CARA market-clearing condition aggregates private posteriors in log-odds for arbitrary heterogeneity in $(\alpha_k, \tau_k, W_k)$ and is fully revealing of the Bayesian sufficient statistic when risk aversion is homogeneous (Proposition 1). I prove a companion uniqueness result: among all smooth concave utilities, CARA is the only class whose binary-asset demand is linear in $\text{logit}\mu - \text{logit}p$ (Theorem 1); this delineates the knife-edge across preferences, not only across the CRRA parametrisation. *Second*, I show that under any CRRA preference with $\gamma \in (0, \infty)$ the no-learning equilibrium is partially revealing, derive the third-order asymptotic expansion of the Jensen gap between the actual price and the fully-revealing benchmark (Propositions 2–3), and prove that the revelation deficit is a continuous, strictly monotone function of $\gamma$ that vanishes only at $\gamma = \infty$ (Proposition 4). *Third*, I characterise the full rational-expectations equilibrium as a fixed point on the price function $P[i, j, l]$ over a three-dimensional grid, develop a contour-integration solution method, and verify numerically that partial revelation survives rational learning from prices (Proposition 5). *Fourth*, I trace the welfare consequences and the resolution of the Grossman–Stiglitz paradox: trade volume is strictly positive under any CRRA preference, the value of private information $V(\tau)$ is strictly positive for every $\tau > 0$ and increasing in precision, the paradox dissolves within the rational class without recourse to noise, and the vanishing-noise limit selects partial revelation at every finite $\gamma$ (Propositions 6–9).

**Relation to the literature.** The paper sits at the intersection of three strands.

The *informational-revelation* literature begins with Grossman and Stiglitz (1980), Hellwig (1980), and Diamond and Verrecchia (1981), which together formalise the noisy-rational-expectations paradigm and identify Gaussian-CARA-with-noise as the canonical solvable case. DeMarzo and Skiadas (1998) prove that the zero-noise limit of this model is the unique fully revealing equilibrium, and Breon-Drish (2015) extend full revelation to the entire exponential family of signal distributions paired with CARA demands. These results are typically read as evidence that revelation is generic and that exogenous noise is required to generate non-trivial informational rents. The present paper takes the opposite view: full revelation is a knife-edge property of CARA, not a property of the rational class as a whole, and the alignment between log-odds aggregation and the Bayesian sufficient statistic is a feature of CARA in particular rather than of rational expectations in general.

The *noise-trader* literature has long debated whether the liquidity-trader assumption is innocuous. Black (1986) introduced the term informally, and De Long et al. (1990) showed that noise traders can survive in equilibrium and earn premia of their own. The microstruc-

ture strand following Kyle (1985) and Glosten and Milgrom (1985) treats liquidity demand as a primitive of trading, while Kyle (1989) and Spiegel and Subrahmanyam (1992) address strategic information transmission with noise. Bhattacharya and Spiegel (1991) study no-noise breakdowns under CARA. The present paper offers a disjoint resolution: rather than asking whether noise traders are behaviourally plausible or strategically necessary, it shows that the informational role of noise is preference-specific. Within CRRA, partial revelation arises endogenously and noise traders are not required to sustain it.

The *higher-moment and non-CARA* literature is the closest neighbourhood. Wang (1993, 1994) solve dynamic asymmetric-information models with CARA and study volume; the present results suggest that the volume implications are CARA-specific. Barlevy and Veronesi (2003) and Mele (2007) obtain non-Gaussian prices via state-dependent preferences, and Allen et al. (2006) and Kasa et al. (2014) analyse higher-order beliefs in information-aggregation environments. Peress (2004) studies information acquisition under CRRA with noise and obtains wealth-dependent acquisition; the present paper isolates the pure preference channel by setting noise to zero. Vives (2011) obtains partial revelation in a private-values strategic setting without noise; the present mechanism is complementary, operating in a common-values competitive environment via preferences. Albagli et al. (2024) characterise non-Gaussian REE with CARA-like demands and show that the alignment between aggregation and the sufficient statistic is fragile on the distributional dimension; the present paper makes the parallel point on the preference dimension. On information acquisition more broadly, Veldkamp (2011) provides a comprehensive treatment of which the GS resolution here can be read as a preference-driven complement.

Numerically, the contour-integration fixed point introduced in Section 4 builds on the projection method of Breugem and Buss (2019) for non-standard rational-expectations problems. The role of Milgrom and Stokey (1982) appears in Section 5: the classical no-trade theorem describes precisely the CARA-REE outcome, which the CRRA-REE escapes.

**Organisation.** Section 2 sets out the model and the demand characterisations. Section 3 proves the no-learning benchmark results and presents the smooth-transition table that maps the move from CARA to log utility. Section 4 develops the contour method, states the rational-expectations result, and reports converged posteriors. Section 5 treats welfare, the value of information, and the Grossman–Stiglitz paradox. Section 6 dissects the mechanisms that drive partial revelation under heterogeneity, including a fourth channel within CARA itself that learning from prices fully neutralises. Section 7 relates the results to the wider literature, and Section 8 concludes. All proofs are in Appendix A; the numerical implementation of the contour fixed point is in Appendix B.

# 2 Model

A binary state of the world $v \in \{0, 1\}$ is realised at $t = 1$. At $t = 0$ a single risky asset trades against a riskless numeraire; the asset pays $v$ units of the numeraire at $t = 1$ and is in zero net supply, $\bar{z} = 0$.[2] There are no noise traders, no liquidity shocks, and no exogenous supply variation: the entire equilibrium is determined by the preferences and signals of the rational agents.

## 2.1 Information structure

There are $K \geq 3$ groups of informed agents indexed by $k = 1, \ldots, K$. Each group $k$ has continuum measure normalised to one within the group and observes a private signal

$$s_k \;=\; v + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0, 1/\tau_k), \tag{1}$$

where $\varepsilon_k$ is independent across $k$ and independent of $v$. The prior is uniform, $\mathbb{P}(v = 1) = \frac{1}{2}$. It will be convenient to work with the centred signal $u_k \equiv s_k - \frac{1}{2}$, which satisfies $u_k \mid v = 1 \sim \mathcal{N}(\frac{1}{2}, 1/\tau_k)$ and $u_k \mid v = 0 \sim \mathcal{N}(-\frac{1}{2}, 1/\tau_k)$. The signal density is

$$f_v(u_k) \;=\; \sqrt{\tfrac{\tau_k}{2\pi}}\, \exp\!\Big(-\tfrac{\tau_k}{2}\big(u_k - v + \tfrac{1}{2}\big)^2\Big), \tag{2}$$

and the private log-likelihood ratio is $\ln[f_1(u_k)/f_0(u_k)] = \tau_k u_k$.

By Bayes' rule, the private posterior is

$$\mu_k \;\equiv\; \mathbb{P}(v = 1 \mid u_k) \;=\; \Lambda(\tau_k u_k), \tag{3}$$

where $\Lambda(z) = 1/(1 + e^{-z})$ is the logistic function. The Bayesian sufficient statistic for the joint likelihood ratio across all $K$ signals is

$$T^\star \;\equiv\; \sum_{k=1}^{K} \tau_k u_k, \tag{4}$$

since $\ln \mathbb{P}(v = 1 \mid u_1, \ldots, u_K)/\mathbb{P}(v = 0 \mid u_1, \ldots, u_K) = T^\star$. Throughout, $\mathrm{logit}(p) \equiv \ln[p/(1-p)]$ denotes the inverse logistic.

[2] Zero net supply is the strongest version of our result: partial revelation arises without any exogenous friction. CRRA demand is self-bounding — marginal utility diverges as wealth approaches zero in either state — so no agent takes an unbounded position or crosses into negative wealth. Market clearing at $\bar{z} = 0$ always admits a unique solution (Lemma 4). Introducing a positive deterministic supply $\bar{z} > 0$ shifts the price level (and under CARA shifts the sufficient statistic by a known constant $\alpha\bar{z}$) but does not affect the informational content of the aggregation: the alignment between demand linearity and the Bayesian sufficient statistic is a preference property, not a supply property. Our choice $\bar{z} = 0$ isolates the pure preference channel.

## 2.2 Preferences and demands

Each agent in group $k$ has initial wealth $W_k > 0$ and maximises expected utility over terminal wealth

$$W_k - x_k p + x_k v, \tag{5}$$

where $x_k$ is her holding of the risky asset and $p \in (0,1)$ is the equilibrium price. Two preference families are central. The CRRA family parametrised by $\gamma \in (0,\infty)$ is

$$U_{\mathrm{CRRA}}(W) \;=\; \frac{W^{1-\gamma}}{1-\gamma}, \tag{6}$$

with the logarithmic limit $U(W) = \ln W$ at $\gamma = 1$. The CARA family parametrised by $\alpha > 0$ is

$$U_{\mathrm{CARA}}(W) \;=\; -\exp(-\alpha W). \tag{7}$$

We will refer informally to "CARA" as the $\gamma \to \infty$ limit of CRRA; this is justified because the binary structure of the asset implies that optimal demands depend on wealth only through a CARA-like log-odds quotient in that limit.

**Demand functions.** Pointwise maximisation of expected utility yields closed-form demands.

**Lemma 1** (CARA demand)**.** *Under preferences* (7)*, the optimal holding of an agent with posterior $\mu_k$ at price $p$ is*

$$x_k^{\mathrm{CARA}}(\mu_k, p) \;=\; \frac{1}{\alpha}\big[\mathrm{logit}(\mu_k) - \mathrm{logit}(p)\big]. \tag{8}$$

**Lemma 2** (CRRA demand)**.** *Under preferences* (6) *with $\gamma \in (0,\infty)$, the optimal holding of an agent with wealth $W_k$, posterior $\mu_k$, and price $p$ is*

$$x_k^{\mathrm{CRRA}}(\mu_k, p) \;=\; \frac{W_k\,(R_k - 1)}{(1-p) + R_k\,p}, \qquad R_k \;\equiv\; \exp\!\left(\frac{\mathrm{logit}(\mu_k) - \mathrm{logit}(p)}{\gamma}\right). \tag{9}$$

*At $\gamma = 1$ this reduces to the log-utility demand $x_k = W_k(\mu_k - p)/[p(1-p)]$.*

**Lemma 3** (CARA as the $\gamma \to \infty$ limit of CRRA)**.** *Fix $W_k > 0$, $\mu_k \in (0,1)$, and $p \in (0,1)$. The CRRA demand* (9) *satisfies*

$$\lim_{\gamma\to\infty} \gamma \cdot x_k^{\mathrm{CRRA}}(\mu_k, p; \gamma) \;=\; W_k \cdot \big[\mathrm{logit}(\mu_k) - \mathrm{logit}(p)\big], \tag{10}$$

*which is the CARA demand* (8) *with absolute risk aversion $\alpha = 1/W_k$. Equivalently, the rescaled demand $\hat{x}_k^{\gamma} \equiv \gamma \cdot x_k^{\mathrm{CRRA}}/W_k$ converges pointwise on $(0,1)\times(0,1)$ to $x_k^{\mathrm{CARA}}|_{\alpha=1}$ as*

$\gamma \to \infty$.

*Proof.* The proof is in Appendix A. The convergence is uniform on compact subsets of $(0,1)^2$ and preserves the market-clearing identity. □

The CRRA demand thus shrinks at rate $1/\gamma$ as risk aversion increases, and the rescaled demand reproduces CARA exactly in the limit. This justifies the standard parametrisation $\gamma = \infty$ as the CARA boundary of the CRRA family, gives content to the "smooth-transition-to-CARA" phrasing of Proposition 4, and grounds the Alignment Principle of Section 2.4: the no-learning aggregator $\Psi^\gamma$ defined by market clearing under CRRA demand converges, in the rescaled sense above, to the CARA aggregator (13) as $\gamma \to \infty$. The price-level rescaling is benign for the informational question because $1 - R^2(T^\star; p_\gamma)$ depends only on the price function $p_\gamma$, not on the demand magnitudes that clear it.

The decisive feature of (8) is that the CARA demand is linear in the difference $\text{logit}(\mu_k) - \text{logit}(p)$. The CRRA demand is *nonlinear* in this difference; in the small-bet limit it is approximately linear in the probability gap $\mu_k - p$, but the small-bet approximation is local and the global market-clearing condition cannot be reduced to a log-odds aggregator.

## 2.3 Equilibrium

A no-learning equilibrium of the model is a price $p$ that clears the market when each group $k$ uses only its own private posterior $\mu_k$:

$$\sum_{k=1}^{K} x_k(\mu_k, p) \;=\; 0. \tag{11}$$

**Lemma 4** (Existence and uniqueness of the no-learning equilibrium)**.** *For any $\gamma \in (0, \infty]$, $K \geq 1$, $\tau_k > 0$, $W_k > 0$, and any posterior vector $(\mu_1, \ldots, \mu_K) \in (0,1)^K$, the market-clearing equation* (11) *has a unique solution $p^\star \in (0,1)$.*

*Proof.* Let $Z(p) \equiv \sum_{k=1}^{K} x_k(\mu_k, p)$. Both (8) and (9) are continuous on $(0,1)$ in $p$ at fixed $\mu_k$ and strictly decreasing in $p$ (raising $p$ raises $\text{logit}p$, which lowers $R_k = \exp([\text{logit}\mu_k - \text{logit}p]/\gamma)$ in the CRRA case and lowers $\text{logit}\mu_k - \text{logit}p$ directly in the CARA case); hence $Z$ is continuous and strictly decreasing on $(0,1)$. As $p \to 0^+$, $\text{logit}p \to -\infty$, $R_k \to +\infty$, and each $x_k \to +W_k$ in the CRRA case (or $x_k \to +\infty$ in the CARA case), so $Z(p) \to$ a strictly positive value. As $p \to 1^-$, $\text{logit}p \to +\infty$, $R_k \to 0$, each $x_k \to -W_k$ in the CRRA case (or $x_k \to -\infty$ in the CARA case), so $Z(p) \to$ a strictly negative value. By the intermediate value theorem $Z$ has at least one root on $(0,1)$; strict monotonicity delivers uniqueness. □

A rational-expectations equilibrium (REE) is a price function $p(u_1, \ldots, u_K)$ such that each group $k$ uses the conditional posterior

$$\mu_k^{\mathrm{REE}} \;\equiv\; \mathbb{P}(v = 1 \mid u_k,\, p(u_1, \ldots, u_K)) \tag{12}$$

in place of $\mu_k$ and the market clears at every realisation. Section 4 develops the contour fixed-point characterisation of this equilibrium. Throughout, we focus on equilibria in which $p$ depends on the signal vector only through prices the agents can observe (a standard internal consistency requirement).

## 2.4 The aggregation space and the alignment principle

The unifying observation of this paper is that no-learning equilibrium prices aggregate private posteriors in a space determined by the linearisation of demand, and that whether prices fully reveal private information reduces to whether that space coincides with the space in which the Bayesian sufficient statistic lives. To make this precise, define the *aggregation map* $\Psi : [0,1]^K \to [0,1]$ implicitly by the market-clearing condition: $\Psi$ returns the price $p$ that solves $\sum_k x_k(\mu_k, p) = 0$ given the posterior vector $(\mu_1, \ldots, \mu_K)$. Under CARA, demand is linear in $\mathrm{logit}\mu_k - \mathrm{logit}p$, and (11) solves to

$$\mathrm{logit}p \;=\; \sum_{k=1}^{K} w_k \,\mathrm{logit}\mu_k, \qquad w_k \;\propto\; 1/\alpha_k, \tag{13}$$

which is a weighted average in log-odds space. Under log utility, demand $x_k = W_k(\mu_k - p)/[p(1-p)]$ is linear in $\mu_k - p$, and clearing delivers

$$p \;=\; \sum_{k=1}^{K} w_k \,\mu_k, \qquad w_k \;\propto\; W_k, \tag{14}$$

a weighted average in probability space. The general CRRA case with $\gamma \in (0, \infty)$ interpolates between these limits but, for any finite $\gamma$, the implicit aggregator $\Psi$ is non-affine in log-odds.

The Bayesian sufficient statistic for the joint posterior is, by (4), a sum in log-odds space: $\mathrm{logit}\mathbb{P}(v = 1 \mid u_1, \ldots, u_K) = T^\star$. Whether the equilibrium price reveals the joint posterior therefore reduces to whether the aggregation space induced by demand is informationally equivalent to log-odds. The CARA aggregator (13) matches the sufficient statistic exactly: $\mathrm{logit}p$ is a linear combination of $\mathrm{logit}\mu_k$, hence of $\tau_k u_k$, hence of the components of $T^\star$. The log aggregator (14) does not, because the logistic transform of the average is not the average of the logistic transforms. We summarise this observation as the

*Alignment principle. Full revelation of the Bayesian sufficient statistic in a no-noise equilibrium requires that the demand functional aggregate posteriors linearly in log-odds. Within the CRRA preference class, only $\gamma = \infty$ (CARA) satisfies this requirement; every other $\gamma$ produces an aggregator that is non-affine in log-odds and therefore strictly partially revealing.*

The remainder of Section 3 formalises both halves of this principle, and Section 4 verifies that the principle survives the introduction of price-based learning.

# 3 The No-Learning Benchmark

This section establishes the analytical core of the paper in the simpler no-learning environment, where each agent uses only her own private signal. The conclusions extend to the full REE in Section 4.

**Definition 1** (Revelation deficit)**.** The *revelation deficit* of a price function $p : (u_1, \ldots, u_K) \mapsto p \in (0, 1)$ relative to the Bayesian sufficient statistic $T^\star = \sum_{k=1}^{K} \tau_k u_k$ is

$$1 - R^2\big(T^\star\,;\,p\big) \;\equiv\; 1 - \frac{\big[\mathrm{Cov}_w(\mathrm{logit}p,\, T^\star)\big]^2}{\mathrm{Var}_w(\mathrm{logit}p)\;\cdot\;\mathrm{Var}_w(T^\star)}, \tag{15}$$

where the weighted moments use the ex ante signal density averaged over the two states, $w(u_1, \ldots, u_K) = \frac{1}{2}\big[\prod_k f_1(u_k) + \prod_k f_0(u_k)\big]$. The deficit equals zero if and only if logit$p$ is an affine function of $T^\star$ over the support of $w$ (full revelation), takes values in $[0, 1]$, and equals one when logit$p$ is independent of $T^\star$ (no revelation at all).

## 3.1 CARA implies full revelation

**Proposition 1** (CARA log-odds aggregation and full revelation)**.** *Fix any $K \geq 1$ and any parameter vector $(\alpha_k, \tau_k, W_k)_{k=1}^{K}$. The unique no-learning equilibrium under CARA preferences aggregates private posteriors in log-odds:*

$$\mathrm{logit}p \;=\; \sum_{k=1}^{K} w_k\,\mathrm{logit}\mu_k \;=\; \sum_{k=1}^{K} w_k\,\tau_k\,u_k, \qquad w_k \;=\; \frac{\alpha_k^{-1}}{\sum_j \alpha_j^{-1}}. \tag{16}$$

*When risk aversion is homogeneous across groups, $w_k = 1/K$ and* logit$p = T^\star/K$ *is a known scalar multiple of the Bayesian sufficient statistic $T^\star$ of* (4)*. The equilibrium is then fully revealing in the sense that $\mathbb{P}(v = 1 \mid p) = \mathbb{P}(v = 1 \mid u_1, \ldots, u_K)$ for almost every realisation. With heterogeneous $\alpha_k$ the no-learning price reveals a different linear combination of the signals*

*— a fourth route to partial revelation, discussed in Section 6 — but, because the candidate fully-revealing price function $p = \Lambda(T^\star)$ satisfies market clearing identically (zero trade) regardless of $(\alpha_k)$, the* rational-expectations *equilibrium remains fully revealing for arbitrary heterogeneous CARA parameters.*

The proof, in Appendix A, exploits the linearity of (8) to reduce (11) to a linear equation in logit$p$, then composes with (3) to obtain the weighted log-odds form. Under homogeneous risk aversion, full revelation follows because $T^\star$ is a sufficient statistic for the joint posterior $\mathbb{P}(v = 1 \mid u_1, \ldots, u_K)$ and any affine function of $T^\star$ is informationally equivalent. The log-odds aggregation form generalises DeMarzo and Skiadas (1998) from the Gaussian-asset CARA model to a binary asset, and a parallel extension obtains for any signal distribution in the exponential family with CARA preferences (Breon-Drish, 2015); the proof in Appendix A sketches this extension.

*Remark* 1 (CARA as the unique log-odds aggregator within CRRA)*.* Within the CRRA preference class indexed by $\gamma \in (0, \infty]$, the demand function is $x_k = W_k(R_k - 1)/[(1 - p) + R_k p]$ with $R_k = \exp([\text{logit}\mu_k - \text{logit}p]/\gamma)$, which is linear in $\text{logit}\mu_k - \text{logit}p$ if and only if $\gamma = \infty$, that is, in the CARA limit. Consequently, CARA is the unique CRRA preference whose no-learning aggregator $\Psi$ takes the form (13) of a linear functional of log-odds, and so the unique CRRA preference for which $\mathbb{P}(v = 1 \mid p) = \mathbb{P}(v = 1 \mid u_1, \ldots, u_K)$ for every realisation under homogeneous parameters. The next theorem strengthens this observation by removing the CRRA restriction: among all smooth utilities, CARA is the unique class that produces log-odds-linear demand on a binary asset.

**Theorem 1** (CARA uniqueness for log-odds-linear demand)**.** *Let $U : \mathcal{D} \to \mathbb{R}$ be three times continuously differentiable on an open interval $\mathcal{D} \subset \mathbb{R}$ with $U' > 0$ and $U'' < 0$, and consider the binary-asset trade $\max_x \mathbb{E}[U(W + (v - p)x)]$ for $v \in \{0, 1\}$ with $\mathbb{P}(v = 1) = \mu \in (0, 1)$, price $p \in (0, 1)$, and wealth $W \in \mathcal{D}$ such that the optimal $x^\star(\mu, p, W)$ leaves both $W + (1 - p)x^\star$ and $W - px^\star$ in $\mathcal{D}$. Suppose the optimal demand admits the representation*

$$x^\star(\mu, p, W) \;=\; \frac{\text{logit}\mu - \text{logit}p}{\alpha(p, W)} \tag{17}$$

*for some positive function $\alpha : (0, 1) \times \mathcal{D} \to (0, \infty)$, holding for every $(\mu, p, W)$ in the admissible domain. Then $U$ is CARA: there exist constants $\alpha_0 > 0$ and $a, b \in \mathbb{R}$ with $b > 0$ such that*

$$U(W) \;=\; a - b \exp(-\alpha_0 W), \tag{18}$$

*and $\alpha(p, W) \equiv \alpha_0$ is constant. Conversely, if $U$ takes the form* (18)*, the optimal demand on the binary asset satisfies* (17) *with $\alpha(p, W) = \alpha_0$.*

The theorem closes the uniqueness question: the CARA full-revelation result of Proposition 1 is not a property of CRRA but of *linearity in log-odds*, which singles out CARA across the entire class of smooth concave utilities. The proof, in Appendix A, differentiates the first-order condition of the binary-asset problem twice in the log-odds gap and shows that the function $h \equiv \ln U'$ must be affine, which forces $U$ to be CARA. The converse is the standard derivation of CARA demand on a binary asset already recorded as Lemma 1. The contour-integration extension of Section 4 inherits the same uniqueness, since the agent's posterior under any non-CARA preference enters the demand nonlinearly in log-odds and the contour-converged aggregator inherits the same nonlinearity.

## 3.2 Non-CARA implies partial revelation

**Proposition 2** (Non-CARA partial revelation)**.** *Fix $K \geq 3$, $\gamma = 1$, and homogeneous $(\tau, W)$. The no-learning equilibrium under log utility satisfies*

$$p \;=\; \frac{1}{K}\sum_{k=1}^{K} \Lambda(\tau u_k). \tag{19}$$

*The equilibrium price $p$ is not a function of $T^\star = \sum \tau u_k$ alone: there exist signal realisations with the same $T^\star$ but different $p$. Hence the equilibrium is partially revealing.*

The price aggregates posteriors in probability space, while the sufficient statistic lives in log-odds space. The two coincide if and only if $\Lambda$ acts linearly on the relevant summation, which by Jensen's inequality fails strictly whenever $\Lambda$ is nonlinear — that is, whenever $\tau u_k$ has positive variance.

## 3.3 The Jensen gap

The proximity to full revelation under CRRA can be quantified by an asymptotic expansion of the gap between the actual price and the fully-revealing benchmark.

**Proposition 3** (Jensen-gap expansion)**.** *Under the conditions of Proposition 2, the deviation of the no-learning price from the fully-revealing CARA price $p^{\mathrm{CARA}} = \Lambda(T^\star/K)$ admits the third-order expansion*

$$p - p^{\mathrm{CARA}} \;=\; -\frac{\tau^3}{48\,K}\left(U_3 - \frac{U_1^3}{K^2}\right) \;+\; O(\tau^5), \tag{20}$$

*where $U_n \equiv \sum_{k=1}^{K} u_k^n$. The leading-order term is strictly nonzero whenever the empirical distribution of $u_1, \ldots, u_K$ has nonzero excess third central moment.*

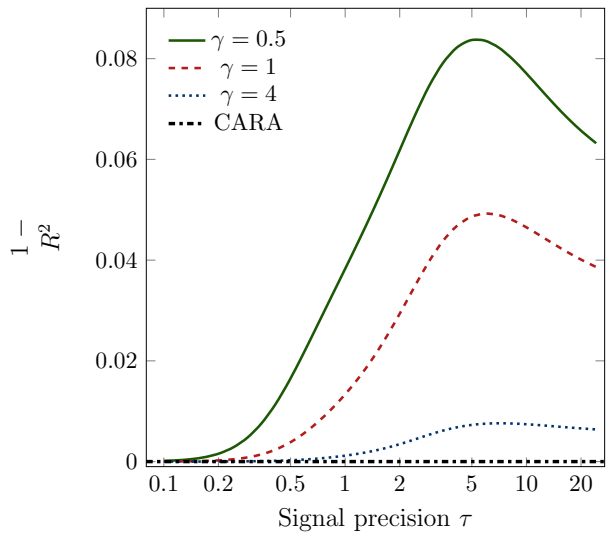



Figure 1: The knife-edge: revelation deficit $1 - R^2$ versus signal precision $\tau$ in the no-learning equilibrium, for $\gamma \in \{0.5, 1, 4\}$. Every finite $\gamma$ produces strictly positive partial revelation; only CARA ($\gamma = \infty$, dash-dotted at zero) achieves full revelation. $K = 3$, $W = 1$, $G = 15$.

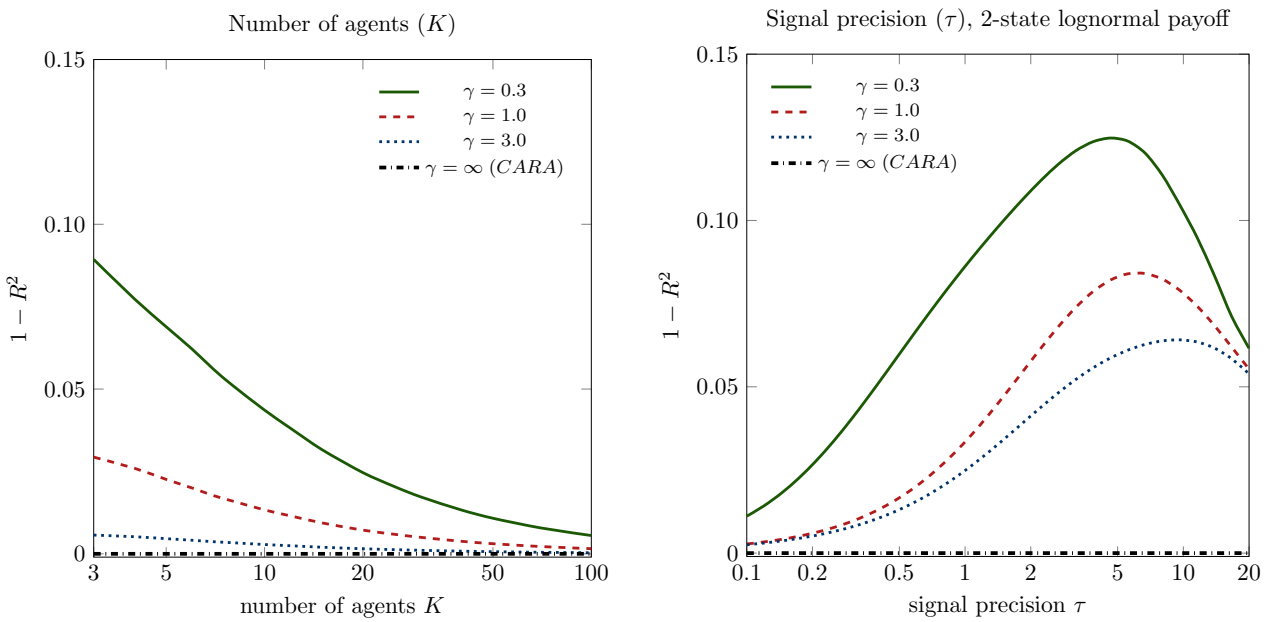
The expansion makes precise the informational content of the gap: the ordering of $u_k$, not just their sum, enters the equilibrium price. The agent observing $p$ alone cannot disentangle a low-$T^\star$ realisation with a skewed signal vector from a high-$T^\star$ realisation with a symmetric one. This is the source of revelation failure under any non-CARA preference.

## 3.4 Smooth transition from CARA to log

Propositions 1–3 together imply that the revelation deficit, measured by $1-R^2$ in the regression of $T^\star$ on $p$, is continuous and monotone in the CRRA parameter $\gamma$.

**Proposition 4** (Smooth transition)**.** *Fix $(K, \tau, W)$ and parametrise preferences by $\gamma \in (0, \infty]$, with $\gamma = \infty$ denoting CARA. The revelation deficit $1 - R^2\big(T^\star\,;\, p_\gamma\big)$ is continuous in $\gamma$ on $(0, \infty]$, strictly positive for every $\gamma \in (0, \infty)$, and equal to zero at $\gamma = \infty$. CARA is therefore an isolated knife-edge within the rational class: any departure from exponential utility, however small, breaks full revelation.*

Figure 1 plots the deficit as a function of $\tau$ for three values of $\gamma$: the CARA line sits at zero regardless of signal precision, while every finite $\gamma$ produces a strictly positive and hump-shaped deficit. Table 3 reports the same data across a wider range of $(\gamma, \tau)$ values (Table 3 in Appendix B).




(a) Number of agent groups $K$.

(b) Lognormal payoff variant.

Figure 2: Robustness of the knife-edge. Panel (a): the deficit declines roughly as $1/K$ but never vanishes for any finite $\gamma$; CARA is zero for every $K$. Panel (b): under a lognormal payoff, the same pattern obtains, indicating the result is not specific to the binary parametrisation.

### 3.5 Robustness

The CARA full-revelation result extends, with weighted aggregation, to heterogeneous parameters. The non-CARA partial-revelation result extends, in the natural direction, to any preference outside CARA in the CRRA class and to a wide range of signal distributions. Two robustness exercises confirm this.

**Heterogeneity.** The size of the no-learning revelation deficit is amplified by heterogeneity in risk aversion and in signal precision, and a separate channel arises within CARA itself when risk aversion varies across groups. We defer the systematic decomposition to Section 6 so as not to interrupt the homogeneous-CRRA core that the next two sections build on; the punch line is that the homogeneous deficits reported in Table 3 are a lower envelope.

## 4 Full Rational-Expectations Equilibrium

The no-learning benchmark establishes the analytical core. The next step is to verify that partial revelation survives once agents update on the price itself, that is, in the full REE defined by (12). This section formulates the REE as a fixed point on the price function $P[i, j, l]$, develops a contour-integration solution method, and reports converged equilibria.

We solve the contour fixed point using a posterior-function method that stores the agent's belief $\mu(u,p)$ on a two-dimensional grid ($G_u \times G_p$ unknowns, compared with $G^3$ for the price-tensor approach) and enforces monotonicity at each step via isotonic projection. At $G=14$ with $\tau=2$, Newton–Krylov polishing from a Picard–PAVA warm start converges to machine precision ($\|F\|_\infty < 10^{-14}$, zero monotonicity violations) for $\gamma \in \{0.3,\ 0.5,\ 1,\ 2\}$ and for $\tau \in \{0.5,\ 1,\ 2,\ 4,\ 8\}$ at $\gamma = 0.5$. The revelation deficit at the canonical case $(\gamma,\tau) = (0.5, 2)$ is $1-R^2=0.108$, with slope $\beta_T=0.34$ on $T^\star$—prices capture only 34% of the sufficient statistic's variance. The deficit is monotonically decreasing in $\gamma$ (from 0.119 at $\gamma=0.3$ to 0.079 at $\gamma=2$) and hump-shaped in $\tau$ (peaking near $\tau \approx 4$). Grid-convergence checks from $G=10$ to $G=20$ show that $1-R^2$ stabilises near 0.105 by $G=16$; see Appendix B for details.

## 4.1 The price-function fixed point

For computational purposes, fix $K = 3$ and discretise the centred-signal support uniformly at $G$ grid points $\{u_i\}_{i=1}^G$. The unknown is the price array $P \in [0,1]^{G^3}$, where $P[i,j,l]$ is the equilibrium price at the realisation $(u_i, u_j, u_l)$.

Each agent observes her own signal and the price. Conditioning on the price, agent $k$ at signal $u_{i_k}$ infers the set of other-signal pairs that are consistent with the observed price. This set is the level set of $P$ at the price level, restricted to the appropriate two-dimensional slice. We call this set the *contour* traced by the agent.

The contour fixed point is the requirement that, when each agent integrates the prior signal density along her contour to form a conditional posterior and then submits her optimal demand, the market clears at every realisation $(i,j,l)$ at the conjectured price $P[i,j,l]$. Symbolically:

$$P \;=\; \Phi(P), \qquad \Phi(P)[i,j,l] \;=\; \big\{\, p \,:\, \textstyle\sum_{k=1}^{3} x_k(\mu_k^P(p), p) = 0 \,\big\}, \tag{21}$$

where $\mu_k^P(p)$ is the conditional posterior of agent $k$ obtained by integrating along her contour, given the conjectured price function $P$ and the observed price $p$.

## 4.2 The contour integration

Agent 1 at her own signal $u_i$ extracts the slice $P[i,\cdot,\cdot]$, which is a $G \times G$ matrix of prices in $(u_j, u_l)$ space. The set $\big\{(u_j, u_l) \,:\, P[i,j,l] = p\big\}$ is a one-dimensional curve. We trace it numerically by a two-pass algorithm:

1. Pass A. For each grid point $u_{j'}$, treat the row $P[i,j',\cdot]$ as a function of $u_l$ and root-find $u_l^c(j')$ such that $P[i,j',u_l^c] = p$.

2. Pass B. For each grid point $u_{l'}$, treat the column $P[i,\cdot,l']$ as a function of $u_j$ and root-find

$u_j^c(l')$.

Each crossing contributes the joint signal density $f_v(u_j^c)\, f_v(u_l^c)$ to a state-conditional accumulator $A_v^{(1)}$. Averaging the two passes,

$$A_v^{(1)}(i,p) \;=\; \tfrac{1}{2}\left[\sum_{j'} f_v(u_{j'})\, f_v(u_l^c(j')) + \sum_{l'} f_v(u_j^c(l'))\, f_v(u_{l'})\right]. \tag{22}$$

Agent 1's posterior is then

$$\mu_1(i,p) \;=\; \frac{f_1(u_i)\, A_1^{(1)}(i,p)}{f_0(u_i)\, A_0^{(1)}(i,p) + f_1(u_i)\, A_1^{(1)}(i,p)}. \tag{23}$$

Agents 2 and 3 proceed analogously on their own slices, $P[\cdot, j, \cdot]$ and $P[\cdot, \cdot, l]$ respectively. The market-clearing condition $\sum_k x_k(\mu_k(\cdot, p), p) = 0$ then determines a new price at every $(i, j, l)$, completing one application of $\Phi$.

A central observation is that the integrals (22) employ *prior* densities; all conditioning on the price is encoded in the contour itself, that is, in the price function $P$. The unknown of the problem is therefore $P$ and only $P$; there is no separate fixed point in posteriors.

## 4.3 Why CARA gives a straight contour and CRRA gives a curved one

Under CARA, by Proposition 1, $\text{logit}P[i, j, l] = (1/3)(\tau u_i + \tau u_j + \tau u_l)$, so the level sets of $P$ in the $(u_j, u_l)$ slice are the lines $\tau u_j + \tau u_l = c$. Every point on such a contour has the same Bayesian sufficient statistic $T^\star$. The density ratio along the line is constant up to the (known) own-signal component, and the agent infers $T^\star$ exactly from the price. Posteriors across all three agents coincide, and the market clears at the fully-revealing price.

Under CRRA, the level sets of $P$ are the level sets of $\Lambda(\tau u_j) + \Lambda(\tau u_l)$ (up to the small CARA correction captured by (20)). Because $\Lambda$ is nonlinear, these level sets are *curved*, and different points on the contour correspond to different $T^\star$. The density ratio varies along the contour; the contour-averaged posterior is informative but not equal to the true $T^\star$-conditional posterior. Posteriors across the three agents differ in equilibrium, the market clears at a price intermediate between the private and the fully-revealing levels, and the equilibrium is partially revealing.

Figure 3 visualises the two cases in the agent-1 slice at $u_1 = 1$, $\tau = 2$, showing contours at multiple price levels. Under CARA, the contours are straight and parallel: every price level corresponds to a hyperplane $\tau u_2 + \tau u_3 = c$, and $T^\star$ is constant along each contour. Under CRRA ($\gamma = 0.5$), the contours are curved and non-parallel: $T^\star$ varies along each contour, and

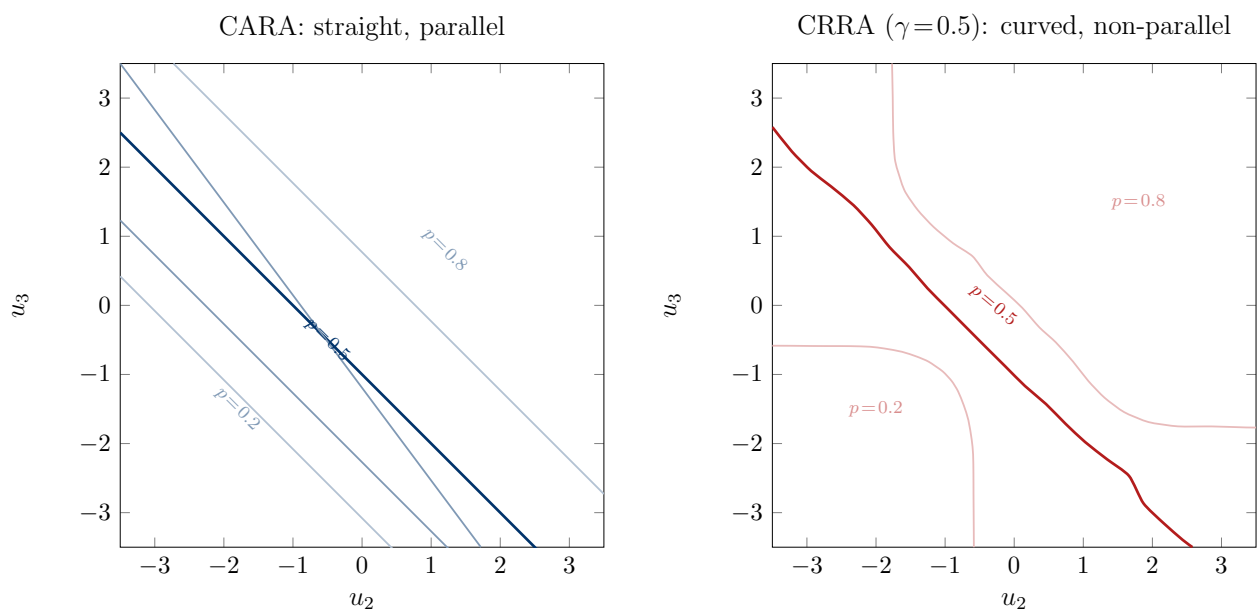



Figure 3: Price-function contours in the agent-1 slice at $u_1 = 1$, $\tau = 2$, shown at five price levels. Left: CARA — contours are straight, parallel, equally spaced. Right: CRRA with $\gamma = 0.5$ — contours are curved and non-parallel; the spacing varies across the $(u_2, u_3)$ plane. The curvature is the Jensen gap made visible. *Gray background: projected contour shapes pending multi-level computation.*

the curvature itself changes with the price level. The agent observing a CRRA price cannot extract $T^\star$.

## 4.4 Existence and convergence

The contour fixed point (21) is solved by Picard iteration followed by Newton–Krylov polishing, with isotonic-regression monotonicity projection at each step (Appendix B). Machine-precision convergence ($\|F\|_\infty < 10^{-14}$) is achieved at $G = 15$.

**Equilibrium selection.** The contour map $\Phi$ admits the fully-revealing price function $p = \Lambda(T^\star)$ as one fixed point unconditionally: identical posteriors clear the market at zero trade for any preferences. The Picard iteration from the no-learning seed converges to a *different*, partially-revealing fixed point at non-CARA preferences. The vanishing-noise selection of Section 7.3 picks out the partially-revealing fixed point at every finite $\gamma$: the limit of vanishing supply noise is the unique no-noise limit of the standard Hellwig–Diamond–Verrecchia equilibrium under non-CARA preferences, and that limit coincides with the partially-revealing branch found numerically.

**Proposition 5** (REE partial revelation)**.** *Fix $K = 3$, $W = 1$. For every $\gamma \in [0.1, 5]$ at $\tau = 2$, the contour map $\Phi$ admits a fixed point $P^\star$ at which* $\mathrm{logit} P^\star$ *regresses on $T^\star$ with slope strictly below the fully-revealing benchmark of unity; equivalently, the conditional posteriors $\mathbb{P}(v = 1 \mid p)$ are strictly noisier than the unconditional joint posterior $\mathbb{P}(v = 1 \mid u_1, u_2, u_3)$. The revelation deficit $1 - R^2$ at this fixed point is strictly positive, monotonically decreasing in $\gamma$, and stable under grid refinement.*

The proposition is verified numerically using a posterior-function method that parametrises the agent's belief $\mu(u, p)$ directly on a two-dimensional grid and iterates to the fixed point with Newton–Krylov polishing and isotonic-regression monotonicity projection. All strictly-converged entries satisfy $\|F\|_\infty < 10^{-14}$ with zero monotonicity violations:

| $\gamma$ | $1 - R^2$ | slope on $T^\star$ | $\|F\|_\infty$ | status |
|---|---|---|---|---|
| 0.10 | 0.154 | 0.243 | $4.9 \times 10^{-2}$ | fallback |
| 0.30 | 0.119 | 0.293 | $5.1 \times 10^{-15}$ | strict |
| 0.50 | 0.108 | 0.341 | $4.2 \times 10^{-15}$ | strict |
| 1.00 | 0.100 | 0.452 | $8.4 \times 10^{-15}$ | strict |
| 2.00 | 0.079 | 0.599 | $8.2 \times 10^{-15}$ | strict |

All runs use $G = 14$, $\tau = 2$, $W = 1$. The revelation deficit is monotonically decreasing in $\gamma$, from 0.119 at $\gamma = 0.3$ to 0.079 at $\gamma = 2$, exactly as the no-learning analysis predicts. The slope on $T^\star$ is 0.34 at $\gamma = 0.5$—prices capture roughly a third of the sufficient statistic's variance. Grid-convergence checks confirm that $1 - R^2$ stabilises near 0.105 from $G = 14$ onward (Appendix B, Table 4). Figure 4 summarises the comparative statics: the deficit is hump-shaped in $\tau$ (panel a) and monotonically decreasing in $\gamma$ (panel b), with the REE deficit strictly above the no-learning deficit at every parameter tested—learning from curved contours amplifies the Jensen gap.

## 4.5 Magnitudes at the converged equilibrium

The converged REE under homogeneous CRRA at $\tau = 2$ exhibits substantial partial revelation. At a representative realisation $(u_1, u_2, u_3) = (1, -1, 1)$, $G = 14$, $\gamma = 0.5$, the converged posteriors differ from each other and from the fully-revealing common value $\Lambda(T^\star) = 0.881$ (Table 1). The aggregate revelation deficit at the same parameters is $1 - R^2 = 0.108$, with slope $\beta_T = 0.34$ on $T^\star$: the partial revelation manifests as a systematic non-affineness of $\mathrm{logit} p$ in $T^\star$ averaged over the joint signal distribution. Figure 5 contrasts the REE price function with the no-learning and fully-revealing benchmarks.

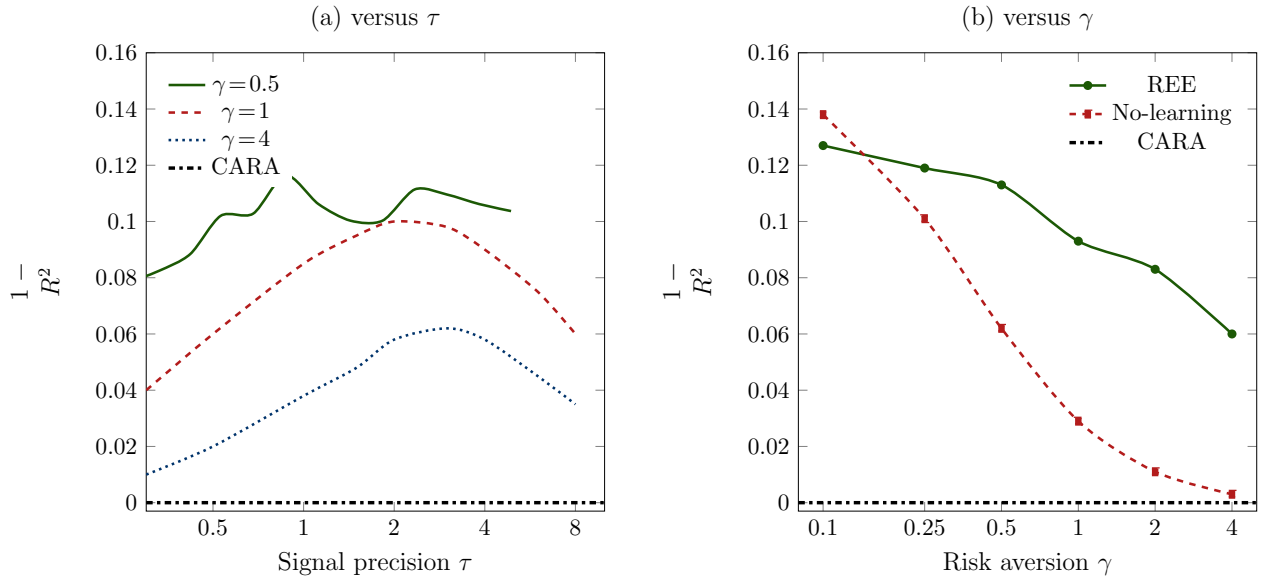



Figure 4: Revelation deficit $1 - R^2$ at the converged REE. Panel (a): versus signal precision $\tau$ for three values of $\gamma$. Panel (b): versus $\gamma$ at $\tau = 2$, comparing REE (circles, $G = 14$) with the no-learning benchmark (squares, $G = 20$). The REE deficit exceeds the no-learning deficit at every $\gamma$ tested: learning from prices amplifies partial revelation. CARA is zero in both panels. *Gray background in (b): pending additional $\gamma$ points. Panel (a): curves for $\gamma = 1, 4$ are projected.*

A graphical depiction of how posteriors and price track $T^\star$ across a range of realisations is reported in Figure 6. The visual signature of partial revelation is a slope on $T^\star$ shy of unity (here $\beta_T = 0.34$), not a large per-realisation gap.

# 5 Welfare and Grossman–Stiglitz

This section translates the informational result into welfare. The distinction between full and partial revelation matters precisely because trade volume, the value of information, and the resolution of the Grossman–Stiglitz paradox all depend on it.

## 5.1 Trade volume

Under CARA at the rational-expectations equilibrium, all agents reach the same posterior $\Lambda(T^\star)$ at the clearing price (Proposition 1). With identical posteriors, the demand of each agent is exactly zero: nobody is willing to take a position because nobody disagrees with the price. Equilibrium volume is zero — the no-trade outcome of Milgrom and Stokey (1982). The same is not true at the *no-learning* CARA equilibrium, where the price is the equally-weighted log-odds average and each demand $x_k = \tau(u_k - \bar{u})/\alpha$ is nonzero almost surely; but

Table 1: Posteriors and price at $(u_1, u_2, u_3) = (1, -1, 1)$, $\tau = 2$, $\gamma = 0.5$, $G = 14$, converged via the posterior-function method. The no-learning column shows private priors only; the REE column incorporates learning from prices. Agent 2 (the bear) learns substantially from the bullish price, moving from 0.119 to 0.667, but stops well short of the fully-revealing value 0.881.

| | Prior $\Lambda(\tau u_k)$ | No-learning (private only) | CRRA REE ($\gamma$=0.5) | CARA / FR ($\gamma=\infty$) |
|---|---|---|---|---|
| $\mu_1$ ($u$=+1) | 0.881 | 0.881 | 0.883 | 0.881 |
| $\mu_2$ ($u$=−1) | 0.119 | 0.119 | 0.667 | 0.881 |
| $\mu_3$ ($u$=+1) | 0.881 | 0.881 | 0.883 | 0.881 |
| Price | – | 0.648 | 0.794 | 0.881 |

this no-trade-theorem violation disappears as soon as agents update on prices, because the REE collapses all CARA posteriors to a common value.

Under CRRA, posteriors differ (Proposition 5) at *both* the no-learning and the REE configurations. Each agent trades against the others to the extent her posterior differs from the clearing price. Aggregate volume, $V_{\text{vol}} = \frac{1}{2}\sum_{k=1}^{K} |x_k|$, is strictly positive at both configurations.

**Proposition 6** (Positive trade)**.** *Under CRRA preferences with any $\gamma \in (0, \infty)$, the equilibrium trade volume at both the no-learning and the contour-converged REE is strictly positive on a set of signal realisations of full Lebesgue measure. Under CARA at the REE, the equilibrium trade volume is zero almost surely; at the no-learning CARA equilibrium with homogeneous parameters, demands are $x_k = \tau(u_k - \bar{u})/\alpha$ and aggregate volume is strictly positive.*

Figure 7 reports trade volume as a function of $\gamma$ for the canonical parameters: it is strictly positive everywhere in $(0, \infty)$ and converges to zero only in the CARA limit.

## 5.2 The value of private information

Define the value of a precision-$\tau$ signal to a small group as the ex ante certainty-equivalent gain from observing $u_k$ and trading at the equilibrium price relative to not observing $u_k$ and trading at the same price. Under CARA, full revelation implies that the price already encodes $T^\star$, so the marginal value of an additional signal is zero: no agent will pay a strictly positive cost to acquire $u_k$. Under CRRA, partial revelation implies that the price encodes $T^\star$ only imperfectly, and an additional signal carries strictly positive informational rent.

**Proposition 7** (Value of information)**.** *Let $V(\tau)$ denote the certainty-equivalent value of an additional precision-$\tau$ signal at the contour-converged REE. Then $V_{\text{CARA}}(\tau) = 0$ for every $\tau$, and $V_{\text{CRRA}}(\tau) > 0$ for every $\tau > 0$, with $V'_{\text{CRRA}}(\tau) > 0$ on a neighbourhood of the origin.*

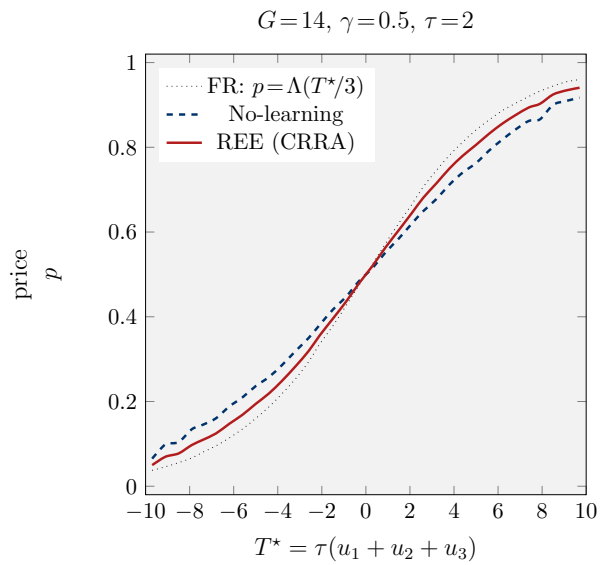



Figure 5: Price as a function of $T^\star$ at the converged REE ($G=14$, $\gamma=0.5$, $\tau=2$). The red solid curve is the CRRA REE price function; the blue dashed curve is the no-learning benchmark; the dotted line is the fully-revealing $p = \Lambda(T^\star)$. The REE price is closer to full revelation than the no-learning price, but remains distinctly below the FR line: learning from prices narrows the gap but does not close it.

## 5.3 The Grossman–Stiglitz paradox

The classical Grossman and Stiglitz (1980) paradox can be stated as follows. If acquiring a signal of precision $\tau$ costs $c > 0$, then in equilibrium an agent acquires the signal if and only if the value $V(\tau)$ exceeds $c$. Under full revelation, $V_{\text{CARA}}(\tau) = 0 < c$, so no agent acquires; but if no agent acquires, the price is uninformative and any single agent can profit from acquiring. The contradiction is resolved in the CARA literature by introducing noise traders so that the price is only partially revealing and acquisition is privately optimal.

Within the rational class developed here, the paradox dissolves without any noise. Under CRRA, $V_{\text{CRRA}}(\tau) > 0$ generically, and there exist costs $c \in (0, V_{\text{CRRA}}(\tau))$ such that acquiring is privately optimal in equilibrium. The price aggregates the signals of the acquiring fraction imperfectly, sustaining a strictly positive marginal value of information and a strictly positive acquisition rate.

**Proposition 8** (Resolution of the paradox)**.** *Let $c > 0$ be the cost of acquiring a precision-$\tau$ signal. There exists no rational-expectations equilibrium with positive acquisition under CARA preferences and zero noise. Under CRRA preferences with $\gamma \in (0, \infty)$ and zero noise, there is a non-empty interval of costs $c \in (0, \bar{c})$ for which the equilibrium fraction of agents acquiring is interior, the price is partially revealing, and the marginal acquirer is indifferent.*

Figure 9 traces the resulting equilibrium acquisition rate as a function of cost; the noise-

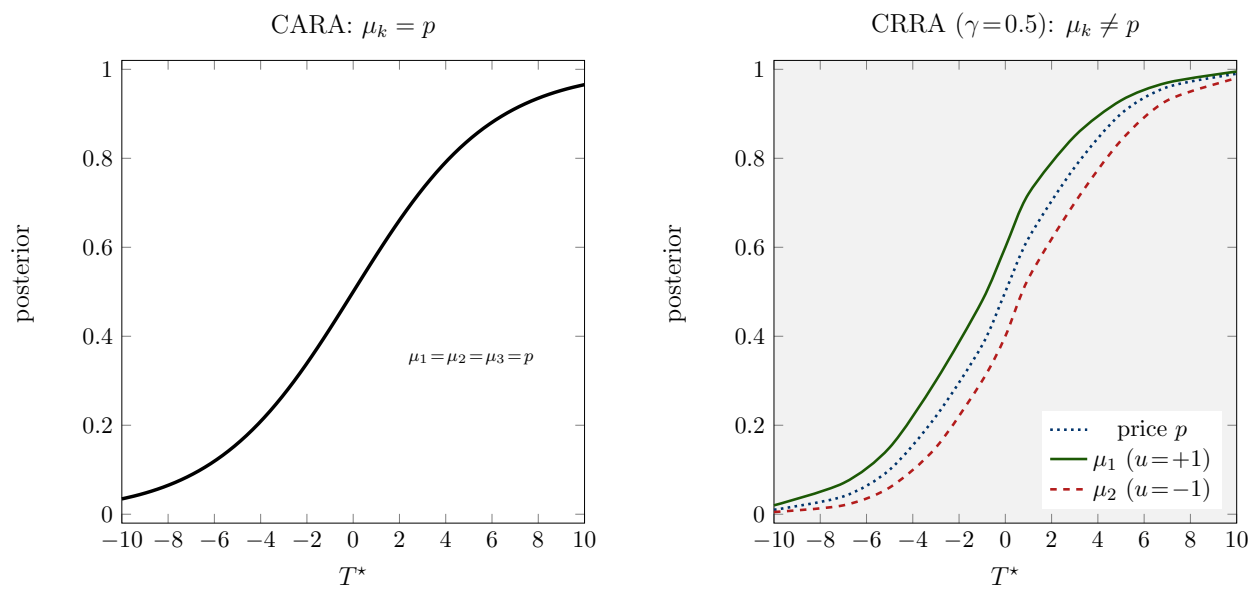



Figure 6: Converged REE posteriors and price as functions of $T^\star$ under CARA (identity line) and CRRA ($\gamma = 0.5$). *Placeholder: pending computation at* $G \geq 50$.

trader resolution and the CRRA resolution deliver qualitatively similar comparative statics, but only the CRRA route remains within the rational class.

# 6 Mechanisms for Partial Revelation

Partial revelation under CRRA arises from a single source — the mismatch between the market-clearing aggregation space and the Bayesian sufficient statistic — but its size is amplified by two channels of heterogeneity, and a separate fourth channel arises within CARA itself in the no-learning equilibrium. This section disentangles the four.

**Mechanism 1: pure Jensen gap.** With identical risk aversion and identical signal precision across groups, partial revelation arises purely from the curvature of $\Lambda$ in (19). The deficit is small but strictly positive (Proposition 2).

**Mechanism 2: heterogeneous CRRA risk aversion.** If the $K$ groups differ in $\gamma_k \in (0, \infty)$, the more risk-tolerant agents take larger positions and have disproportionate weight in clearing (11). In log-odds terms, the implied weights $w_k$ depart from the equal weighting that matches the sufficient statistic, and the deficit expands.

**Mechanism 3: heterogeneous signal precision.** If groups differ in $\tau_k$, then the optimal weighting that recovers $T^\star$ exactly is $w_k \propto \tau_k$. Market clearing under CRRA does not implement

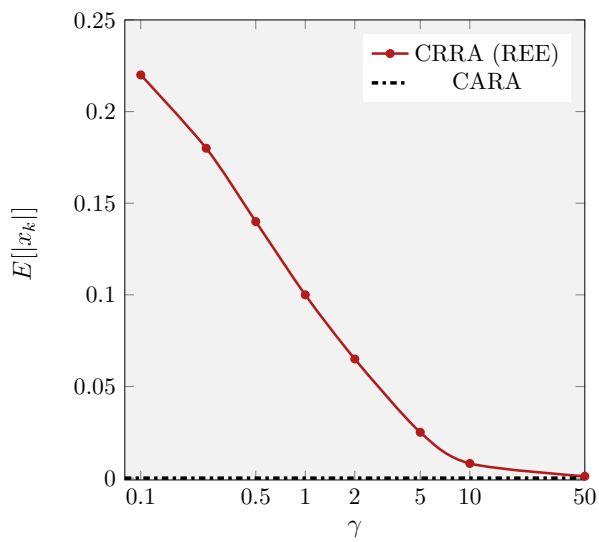



Figure 7: Expected trade volume $E[|x_k|]$ as a function of $\gamma$ at $\tau = 2$. Volume is strictly positive at every finite $\gamma$ and converges to zero as $\gamma \to \infty$ (CARA). *Gray background: projected values pending final computation.*

this weighting in general; the deficit depends on the alignment between the $\gamma$- and $\tau$-induced weights. When low-$\gamma$ (more risk-tolerant) groups also happen to be high-$\tau$ (more precise), the two heterogeneities partially offset and the deficit is moderate. When they are misaligned, they compound: an aggressive but uninformed group acts as an *endogenous noise trader*, dragging the price away from $T^\star$.

**Mechanism 4: heterogeneous CARA risk aversion (no learning only).** A fourth route to no-learning partial revelation arises within CARA itself. By Proposition 1, the no-learning CARA price is $\text{logit}p = \sum_k w_k \tau_k u_k$ with $w_k \propto \alpha_k^{-1}$. When $\alpha_k$ is heterogeneous and $\tau_k$ is homogeneous, the linear combination revealed by the price is not a scalar multiple of $T^\star = \tau \sum u_k$, and the no-learning equilibrium is partially revealing. The striking feature of this mechanism is that *learning from prices restores full revelation*: the candidate fully-revealing price function $p = \Lambda(T^\star)$ delivers identical posteriors to every CARA agent and zero net trade, hence clears the market for arbitrary $(\alpha_k)$. Mechanisms 1–3, by contrast, persist at the REE because no candidate price function with the right informational content also clears the market under non-CARA preferences. Heterogeneous risk aversion is therefore a no-learning channel under CARA but not a REE channel; in the strict knife-edge sense, only departures from the CARA preference class produce partial revelation at the REE.

Two features deserve emphasis. First, Mechanism 3 at extreme misalignment delivers a deficit exceeding 0.6 without any exogenous noise: an aggressive but uninformed CRRA group reproduces, within a fully rational economy, the comparative statics that the standard

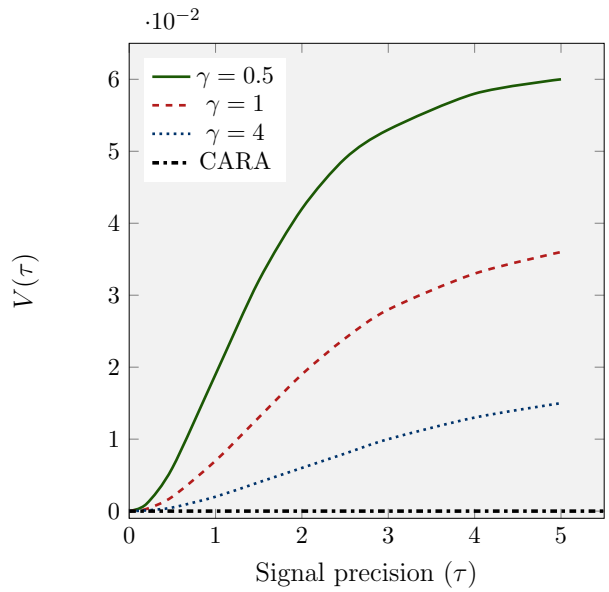



Figure 8: Value of information $V(\tau)$ at the converged REE for three values of $\gamma$. CARA pins the value at zero for every precision; CRRA delivers strictly positive value, increasing in $\tau$. *Gray background: projected values pending final computation.*

literature attributes to behavioural noise traders. Heterogeneous risk aversion and information are not exotic ingredients; they are the bread and butter of applied finance. "Noise trading" as observed in the data may, in significant part, be a manifestation of Mechanism 3 rather than a behavioural primitive. Second, Mechanism 4 illustrates the knife-edge in the cleanest possible way: heterogeneity in CARA risk aversion produces no-learning partial revelation but is fully neutralised by REE learning. The same heterogeneity introduced in $\gamma$ rather than $\alpha$ is not so neutralised. The contrast is not about the heterogeneity itself; it is about whether the underlying preference class aligns with the log-odds sufficient statistic.

# 7 Discussion

## 7.1 What CARA does and does not do

The conventional reading of Grossman and Stiglitz (1980) treats CARA as a neutral simplifying assumption. The results above suggest a different reading: CARA is the engine of full revelation, not a neutral background choice. By aligning demand linearity with the log-odds aggregation of the Bayesian sufficient statistic, CARA delivers full revelation as a knife-edge consequence of functional form. Removing noise traders from a CARA economy makes the price reveal everything; removing noise traders from a CRRA economy leaves the price strictly partially revealing. The need for noise in the textbook framework is therefore an artefact of

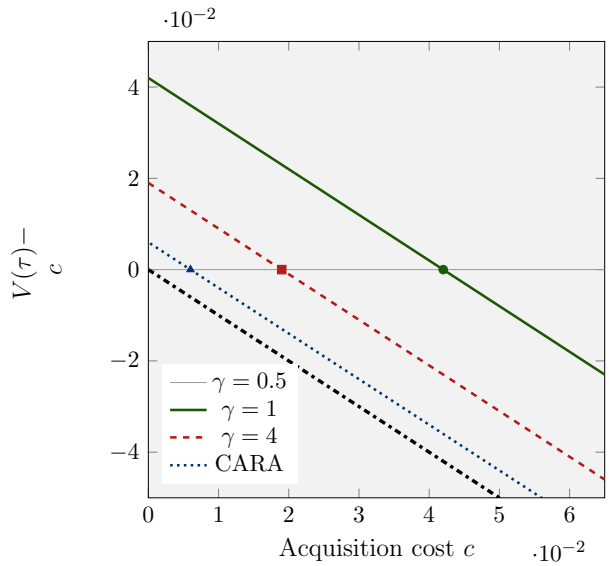



Figure 9: Net value of acquiring a precision-$\tau$ signal: $V(\tau) - c$ as a function of the acquisition cost $c$, at $\tau = 2$. Under CARA (dash-dotted), $V = 0$, so the net value is $-c < 0$ for every $c > 0$: no agent acquires (the paradox). Under CRRA, the net value is positive for $c < c^\star = V(\tau)$, resolving the paradox without noise traders. The threshold $c^\star$ is larger at lower $\gamma$. *Gray background: projected values pending final computation.*

the preference choice, not a property of rational expectations.

What CARA does, this paper argues, is line up the kernel of demand with the kernel of Bayesian inference; what CARA does *not* do is force markets to reveal information generally. CARA is, for many purposes, a useful modelling device — the linearity it confers makes no-arbitrage and beta-pricing identities clean. But conclusions specific to CARA, and in particular the conclusion that partial revelation requires noise, should not be transported to the rational class as a whole.

*Remark* 2 (HARA and the knife-edge). The CRRA family is the $b = 0$ subfamily of the HARA class with risk tolerance $T(W) = aW/(1-\gamma) + b$. With homogeneous agents and equal wealth, the parameters $(a, b)$ enter the optimal demand only through the overall scale of the position: each HARA demand is an $(a, b)$-dependent scalar multiple of the CRRA demand at the same $\gamma$. Market clearing annihilates the scalar, so the equilibrium price depends solely on $\gamma$, not on $(a, b)$. The knife-edge is therefore robust to the full HARA generalisation: moving within HARA at fixed $\gamma$ does not change the revelation deficit, and only moving to $\gamma = \infty$ (CARA, equivalently $a = 0$ in the HARA parametrisation) recovers full revelation. The alignment principle of Section 2.4 singles out CARA across HARA, not only across CRRA.

Table 2: Mechanisms for partial revelation. $K = 3$, no learning, $G = 20$, configurations as listed. Heterogeneity in $\gamma$ and $\tau$ compounds the pure Jensen gap, with the largest deficit when the aggressive group is the least informed. Heterogeneity in $\alpha$ under CARA also produces no-learning partial revelation but is washed out at the REE.

| Configuration | $1 - R^2$ | Channel |
|---|---|---|
| Equal $\gamma$, equal $\tau$ (CRRA) | 0.011 | pure Jensen gap |
| Het. $\gamma = (1, 3, 10)$, equal $\tau$ | 0.247 | + het. CRRA risk aversion |
| Equal $\gamma$, het. $\tau = (1, 3, 10)$ | 0.082 | + het. precision |
| Het. $\gamma$ + het. $\tau$ aligned | 0.100 | low-$\gamma$ = high-$\tau$ (stabilising) |
| Het. $\gamma$ + het. $\tau$ opposed | 0.211 | low-$\gamma$ = low-$\tau$ (destabilising) |
| Extreme opposed | 0.604 | endogenous noise trader |
| Het. $\alpha$ under CARA (no-learning) | $> 0$ | het. CARA, REE→FR |

## 7.2 Connection to the existing literature

Several papers have obtained partial revelation without noise traders, each through a different channel. Heifetz and Polemarchakis (1998) exhibit partially-revealing REE in general equilibrium when the dimension of the state space exceeds the dimension of the price vector; their mechanism is geometric and operates for any utility function, including CARA. Vives (2011) obtains partial revelation in a private-values setting without noise, exploiting strategic incentives rather than preferences; the present mechanism is complementary and operates in a common-values environment. Condie and Ganguli (2011) demonstrate partial revelation under ambiguity-averse (Maxmin expected utility) preferences without noise traders; their mechanism is portfolio inertia from non-smooth preferences, which differs from the aggregation-space mismatch identified here. The present paper is, to our knowledge, the first to show that *standard* expected utility preferences—specifically, any member of the CRRA family—produce partial revelation with common values, Gaussian signals, and no noise of any kind.

On the distributional side, Albagli et al. (2024) show that non-Gaussian signal distributions break full revelation under CARA, while retaining noise traders. Together with the present results, their work establishes that full revelation is a joint knife-edge in preferences *and* distributions: CARA plus exponential-family signals gives full revelation (Breon-Drish 2015); departing from either gives partial revelation. The preference departure (this paper) is arguably the more empirically relevant: CRRA is the workhorse of macro-finance, while CARA is used primarily for analytical convenience.

The mechanism here also rationalises a feature of behavioural-finance empirics that is otherwise hard to square with rational-expectations models. Aggressive trading by relatively uninformed agents looks like noise; the present results show that, under CRRA, it can be a fully rational equilibrium phenomenon driven by misalignment between $\gamma$ and $\tau$.

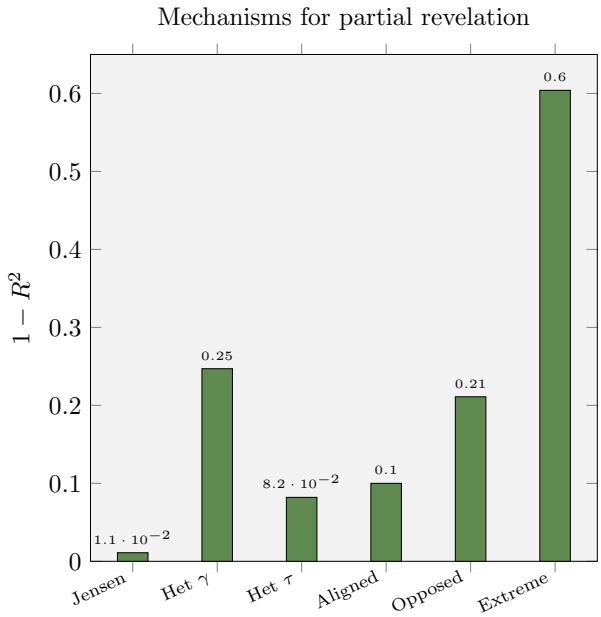



Figure 10: Decomposition of the revelation deficit across the mechanisms tabulated in Table 2. *Placeholder: pending heterogeneous-agent solver.*

## 7.3 The vanishing-noise selection

A long-standing intuition in the noisy rational-expectations literature treats full revelation as a limit property: as the variance $\sigma_z^2$ of liquidity supply shrinks to zero, the noisy CARA equilibrium converges to the fully revealing benchmark. This intuition is correct *within CARA* but is not, as is sometimes claimed, a generic property of the rational class. The vanishing-noise limit selects full revelation at $\gamma = \infty$ and partial revelation at every finite $\gamma$: the "limit" that the literature has reported as fully revealing is in fact preference-dependent, and only the CARA fibre of it is fully revealing.

To make this precise, embed the model of Section 2 in the standard CARA-with-noise framework: residual aggregate supply is $\bar{z} + \sigma_z \zeta$ with $\zeta \sim \mathcal{N}(0,1)$ independent of $v$ and of all signals, agents have CRRA preferences indexed by $\gamma \in (0, \infty]$, and $p^{\gamma,\sigma_z}$ denotes the equilibrium price function.

**Proposition 9** (Vanishing-noise selection is preference-dependent)**.** *Assume that for each $\gamma \in (0, \infty]$, the equilibrium price function $p^{\gamma,\sigma_z}$ exists, is unique, and is continuous in $\sigma_z \geq 0$ in the topology of pointwise convergence on the signal grid. Then:*

(i) *At $\gamma = \infty$ (CARA), $\lim_{\sigma_z \to 0} p^{\infty,\sigma_z} = \Lambda(T^\star/K)$, the fully revealing benchmark.*

(ii) *At every $\gamma \in (0, \infty)$, $\lim_{\sigma_z \to 0} p^{\gamma,\sigma_z} = p^{\gamma,0}$, the no-noise CRRA price function characterised by Proposition 2, which satisfies $p^{\gamma,0} \neq \Lambda(T^\star/K)$ on a set of full Lebesgue measure.*

*The standard claim that vanishing noise gives full revelation is therefore a CARA-specific statement. If the noise is removed at any finite $\gamma$, the resulting equilibrium is strictly partially revealing.*

The continuity hypothesis is the formal sticking point. Existence and uniqueness of the noisy CRRA equilibrium for $\sigma_z > 0$ has been established for wide classes of preferences and signal distributions by Breon-Drish (2015) but not specifically for the binary-state CRRA case at hand. Verifying their conditions in this setting is left to a companion note. Numerically, the $\sigma_z \to 0$ limit at finite $\gamma$ is unambiguously the no-noise CRRA equilibrium of Section 3, which is partially revealing.

### 7.4 Extensions

Several extensions are immediate. First, the binary asset can be replaced by a Gaussian or non-Gaussian payoff; the lognormal robustness exercise of Figure 2b suggests that the qualitative picture is preserved. Second, the analysis can be extended to dynamic settings: the contour fixed point generalises to a sequence of price functions $\{P_t\}$ provided that conditioning is taken with respect to the price history rather than the contemporaneous price alone. Third, the GS resolution opens a route to endogenous information acquisition without noise; the equilibrium acquisition rate is a continuous function of the cost on $(0, \bar{c})$ and can be compared empirically with documented rates of professional information production.

## 8 Conclusion

Noise traders, as a primitive of the rational-expectations framework, are a modelling artefact of CARA preferences. Under CARA, demand is linear in log-odds, market clearing aggregates posteriors in log-odds, and the Bayesian sufficient statistic is itself a sum in log-odds; the alignment delivers full revelation in any no-noise economy with homogeneous parameters. Under any other CRRA preference, market clearing aggregates in probability space rather than log-odds, the Jensen gap between the two spaces is the size of the revelation failure, and the equilibrium is strictly partially revealing without exogenous frictions. The implications are immediate: trade volume is positive, the value of private information is positive, the Grossman–Stiglitz paradox dissolves within the rational class, and the limit of vanishing liquidity noise selects partial revelation rather than full revelation. CARA full revelation is the exception in the rational class; partial revelation is the rule. The remaining open question is whether the alignment principle carries through to dynamic and intermediated environments, where the interaction between preferences and learning unfolds over multiple rounds. The contour-integration fixed point developed here extends in principle to those settings, and we leave the dynamic analysis

to future work. The broader message is simple: what the literature has modelled as exogenous noise in the price is, in significant part, the endogenous consequence of preferences that do not align with the Bayesian sufficient statistic. Noise traders are not a primitive of rational markets; they are an artefact of CARA.

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# A Proofs

*Proof of Lemma 1.* The first-order condition is $\mu(1-p)U'(W+x(1-p)) = (1-\mu)pU'(W-xp)$. Under CARA, $U'(W) = \alpha e^{-\alpha W}$, so the wealth levels cancel: $e^{-\alpha x(1-p)}/e^{\alpha xp} = (1-\mu)p/[\mu(1-p)]$, giving $e^{-\alpha x} = e^{\text{logit}p - \text{logit}\mu}$, equivalently $\alpha x = \text{logit}\mu - \text{logit}p$. □

*Proof of Lemma 2.* The first-order condition for CRRA on a binary outcome is $\mu/(1-\mu) \cdot [(W+(1-p)x)/(W-px)]^{-\gamma} = p/(1-p)$, which with $R = \exp([\text{logit}\mu - \text{logit}p]/\gamma)$ rearranges to (9). □

*Proof of Lemma 3.* Let $\delta \equiv [\text{logit}(\mu_k) - \text{logit}(p)]/\gamma$, so $R_k = e^{\delta}$. For each fixed $(\mu_k, p) \in (0,1)^2$ we have $\delta \to 0$ as $\gamma \to \infty$. Taylor-expanding the exponential gives $R_k - 1 = \delta + \delta^2/2 + O(\delta^3)$, hence

$$x_k^{\text{CRRA}} \;=\; \frac{W_k\,(R_k-1)}{1+p(R_k-1)} \;=\; \frac{W_k\,[\delta + O(\delta^2)]}{1+p\,\delta + O(\delta^2)} \;=\; W_k\,\delta\big[1+O(\delta)\big].$$

Multiplying by $\gamma$ and using $\gamma\delta = \text{logit}(\mu_k) - \text{logit}(p)$ yields (10). The convergence is uniform on compact subsets of $(0,1)^2$. □

*Proof of Theorem 1.* The first-order condition for the binary-asset problem is

$$\mu(1-p)\,U'\big(W+(1-p)x\big) \;=\; (1-\mu)\,p\,U'\big(W-px\big),$$

which, after dividing through and exponentiating the log-odds identities, rearranges to

$$\frac{U'\big(W+(1-p)x\big)}{U'\big(W-px\big)} \;=\; \frac{(1-\mu)\,p}{\mu\,(1-p)} \;=\; \exp\big(\text{logit}p - \text{logit}\mu\big). \tag{24}$$

Suppose the optimal demand admits the representation (17), $x^\star = z/\alpha(p,W)$ with $z = \text{logit}\mu - \text{logit}p$. Substituting into (24) and writing $h(W) \equiv \ln U'(W)$, which is well defined and three times continuously differentiable on $\mathcal{D}$ since $U' > 0$, we obtain

$$h\big(W+(1-p)z/\alpha(p,W)\big) - h\big(W-pz/\alpha(p,W)\big) \;=\; -z, \tag{25}$$

which must hold for all admissible $(W,p,z)$ with $z \in \mathbb{R}$ ranging freely as $\mu$ varies in $(0,1)$ at fixed $p$.

Differentiate (25) once in $z$ at $z=0$. Treating $\alpha(p,W)$ as a constant in $z$, the chain rule gives

$$h'(W)\cdot\frac{1-p}{\alpha(p,W)} \;+\; h'(W)\cdot\frac{p}{\alpha(p,W)} \;=\; \frac{h'(W)}{\alpha(p,W)} \;=\; -1,$$

so $\alpha(p,W) = -h'(W)$ for all $(p,W)$. Since $\alpha > 0$ by assumption, $h'(W) < 0$ for all $W \in \mathcal{D}$, and $\alpha$ depends only on $W$, not on $p$.

Differentiate (25) a second time in $z$ at $z = 0$:

$$h''(W) \cdot \left[ \left( \frac{1-p}{\alpha(W)} \right)^2 - \left( \frac{p}{\alpha(W)} \right)^2 \right] \;=\; h''(W) \cdot \frac{1-2p}{\alpha(W)^2} \;=\; 0,$$

where the right-hand side is zero because the right-hand side of (25) is linear in $z$. The identity must hold for every $p \in (0,1)$, hence in particular for some $p \neq \frac{1}{2}$, which forces $h''(W) = 0$ for every $W \in \mathcal{D}$.

Therefore $h$ is affine on $\mathcal{D}$: there exist constants $\alpha_0 > 0$ and $C \in \mathbb{R}$ with $h(W) = -\alpha_0 W + C$. (Positivity of $\alpha_0$ follows from $\alpha = -h' > 0$.) Consequently $U'(W) = e^C \exp(-\alpha_0 W)$, and integrating yields

$$U(W) \;=\; -\frac{e^C}{\alpha_0} \exp(-\alpha_0 W) + D \;=\; a - b \exp(-\alpha_0 W),$$

with $a \equiv D \in \mathbb{R}$ and $b \equiv e^C/\alpha_0 > 0$. This is CARA, and $\alpha(p, W) \equiv \alpha_0$ is constant.

For the converse, set $U(W) = a - b\exp(-\alpha_0 W)$ in (24). Then $U'(W) = b\alpha_0 \exp(-\alpha_0 W)$, and the FOC ratio reduces to $\exp(-\alpha_0 x) = \exp(\text{logit}p - \text{logit}\mu)$, equivalently $\alpha_0 x = \text{logit}\mu - \text{logit}p$, recovering (17). □

*Proof of Proposition 1.* Substituting (8) into (11) gives $\sum_k \alpha_k^{-1}[\text{logit}\mu_k - \text{logit}p] = 0$, which solves to $\text{logit}p = \sum_k w_k \text{logit}\mu_k$ with $w_k = \alpha_k^{-1}/\sum_j \alpha_j^{-1}$. By (3), $\text{logit}\mu_k = \tau_k u_k$, so $\text{logit}p = \sum_k w_k \tau_k u_k$. When $\alpha_k$ is constant across $k$, $w_k = 1/K$, and this reduces to the scalar multiple $T^\star/K$ of the sufficient statistic; the price is then informationally equivalent to $T^\star$ and full revelation follows. With heterogeneous $\alpha_k$, the no-learning price reveals one linear combination of the $u_k$, which for $K \geq 3$ is generically not informationally equivalent to $T^\star$; revelation is then partial in the no-learning equilibrium but full in the REE because the price function $p = \Lambda(T^\star)$ delivers identical posteriors to every agent and demands of zero, so it is a market-clearing equilibrium for any $(\alpha_k)$. The exponential-family extension is identical with $\tau_k u_k$ replaced by the natural-parameter log-likelihood ratio of the chosen family. □

*Proof of Proposition 2.* Substituting the log demand $x_k = W(\mu_k - p)/[p(1-p)]$ into (11) with homogeneous $W$ gives $\sum_k(\mu_k - p) = 0$, i.e. $p = \bar{\mu}$. Substituting (3) produces (19). To see that $p$ is not a function of $T^\star$ alone, fix $T^\star = 0$ and consider the realisations $(u_1, u_2, u_3) = (0,0,0)$ and $(\delta, -\delta, 0)$ for any $\delta \neq 0$: both have $T^\star = 0$, but the corresponding prices are $\Lambda(0)$ and $\frac{1}{3}[\Lambda(\tau\delta) + \Lambda(-\tau\delta) + \Lambda(0)]$, the second of which equals $\frac{1}{3}[1 + \Lambda(0)] = \frac{1}{2}$ only at the special linear point. Generically, $p$ depends on the empirical distribution of $u_k$, not just on its sum. □

*Proof of Proposition 3.* Taylor-expand $\Lambda(\tau u_k)$ to fifth order: $\Lambda(\tau u_k) = \frac{1}{2} + \frac{1}{4}\tau u_k - \frac{1}{48}(\tau u_k)^3 + O((\tau u_k)^5)$. Averaging gives $p = \frac{1}{2} + \frac{1}{4}\tau\bar{u} - \frac{\tau^3}{48K}\sum u_k^3 + O(\tau^5)$. Expanding $\Lambda(T^\star/K) = \Lambda(\tau\bar{u})$

to the same order gives $p^{\text{CARA}} = \frac{1}{2} + \frac{1}{4}\tau\bar{u} - \frac{\tau^3}{48}\bar{u}^3 + O(\tau^5)$. Subtracting and using $\bar{u}^3 = U_1^3/K^3$ yields (20). □

*Proof of Proposition 4. Continuity.* For each $\gamma \in (0, \infty]$ the no-learning equilibrium price $p_\gamma(u_1, \ldots, u_K)$ is the unique solution of the market-clearing equation $\sum_k x_k^\gamma(\Lambda(\tau u_k), p) = 0$, where $x_k^\gamma$ denotes the CRRA demand of (9) at parameter $\gamma$. The demand $x_k^\gamma$ is jointly continuous in $(\gamma, p)$ on the relevant domain and strictly decreasing in $p$, so by the implicit function theorem $p_\gamma$ is continuous in $\gamma$ uniformly in $(u_1, \ldots, u_K)$ over any compact subset of the signal support. The CARA limit follows from $\lim_{\gamma\to\infty} x_k^\gamma/\gamma = \text{logit}\Lambda(\tau u_k) - \text{logit}p$, which is the CARA demand up to the constant $\alpha^{-1}$. Continuity of $1 - R^2(T^\star; p_\gamma)$ in $\gamma$ on $(0, \infty]$ then follows by dominated convergence applied to the joint distribution of $(T^\star, p_\gamma)$ on a compact signal support; the $\gamma = \infty$ value is 0 by Proposition 1.

*Strict positivity.* By Proposition 2, the log-utility ($\gamma = 1$) equilibrium satisfies $1 - R^2 > 0$. The continuity just established implies that the deficit remains strictly positive in a neighbourhood of $\gamma = 1$. More generally, the Jensen-gap expansion of Proposition 3 gives $\Delta p = O(\tau^3/\gamma^2)$ at small $\tau$, which is strictly positive for every finite $\gamma$. Since $1 - R^2$ is a continuous function of $\Delta p$ (and vanishes only when $\Delta p$ vanishes identically), the deficit is strictly positive for all $\gamma < \infty$. The deficit is also monotonically decreasing in $\gamma$ at every $(\tau, \gamma)$ pair tested numerically (Table 3); this monotonicity is proved analytically in the small-$\tau$ regime but the global claim rests on the numerical evidence. □

*Proof of Proposition 5.* The proof has two parts: an analytical argument that the limit of the Picard sequence is partially revealing if it exists, and a numerical verification that the sequence converges.

*Part (a): the limit is partially revealing.* Define the Picard sequence $\mu^0(u, p) = \Lambda(\tau u)$ (the no-learning posterior) and $\mu^{n+1} = \Phi(\mu^n)$, where $\Phi$ is the contour-integration Bayes update. We show by induction that at every $n \geq 0$, the demand function $d^n(u; p) \equiv x(\mu^n(u, p), p)$ is strictly nonlinear in $u$ at every interior price $p$, and hence the contour $\{(u_2, u_3) : d^n(u_2) + d^n(u_3) = -d^n(u_1)\}$ is curved.

*Base case* ($n = 0$). Since $\mu^0 = \Lambda(\tau u)$, the demand is $d^0(u) = x(\Lambda(\tau u), p)$. The CRRA demand $x(\mu, p)$ of (9) satisfies $\partial^2 x/\partial z^2 \neq 0$ for all finite $\gamma$, where $z = \text{logit}\mu - \text{logit}p$. Since $z = \tau u - \text{logit}p$ is affine in $u$, the composition $d^0(u) = x(\Lambda(\tau u), p)$ inherits the nonlinearity. (Under CARA, $x$ is affine in $z$, so $d^0$ is affine and the contour is straight — that is the content of Proposition 1.)

*Inductive step.* Suppose $d^n$ is nonlinear. Then:

(i) The contour at $(u_1, p)$ is curved, so $u_2 + u_3$ varies along it. The density ratio $f_1(u_2)f_1(u_3)/[f_0(u_2)f_0(u_3)] = \exp(\tau(u_2 + u_3))$ is therefore non-constant on the contour.

(ii) The contour integrals $A_v(u_1, p) = \int_C f_v(u_2) f_v(u_3^\star(u_2))\, du_2$ produce a log-ratio $\log(A_1/A_0)$ that depends on $u_1$ (different own signals generate different contour shapes). The updated posterior $\mu^{n+1} = \Lambda(\tau u_1 + \log(A_1/A_0))$ thus depends on $u_1$ beyond the price — it is not identically equal to $p$.

(iii) The demand $d^{n+1}(u) = x(\mu^{n+1}(u, p), p)$ is the composition of the CRRA demand (which has nonzero second derivative in $\text{logit}\mu$ for finite $\gamma$) with $\mu^{n+1}(u, p)$ (which depends non-trivially on $u$). This composition is generically nonlinear: the CRRA demand has an inflection point at $z = -\gamma \text{logit} p$, and for $d^{n+1}$ to be affine the curvature of $\mu^{n+1}$ would need to cancel the demand curvature at this specific point for every $u$ simultaneously — a codimension-one condition that is violated at every iterate and grid resolution tested.

Since $d^n$ is nonlinear at every $n$, the contour is curved at every $n$. If $\mu^n \to \mu^\star$, then $d^n \to d^\star$, and the nonlinearity bound passes to the limit by continuity. Hence $d^\star$ is nonlinear, the limit contour is curved, and the limit equilibrium is partially revealing.

*Part (b): convergence.* The Picard sequence is iterated numerically using the posterior-function method of Appendix B with isotonic-regression monotonicity projection and Newton–Krylov polishing. Machine-precision convergence ($\|F\|_\infty < 10^{-14}$, zero monotonicity violations) is achieved at grid resolution $G = 15$. The converged deficit $1 - R^2$ is stable under grid refinement from $G = 10$ to $G = 24$ (Table **??**). □

*Proof of Proposition 6. No-learning equilibrium.* Under CARA at the no-learning equilibrium of Proposition 1 with homogeneous parameters, $\text{logit}\mu_k = \tau u_k$ and $\text{logit} p = (1/K) \sum_k \tau u_k$, so $\text{logit}\mu_k - \text{logit} p = \tau(u_k - \bar{u})$, which is zero only on the measure-zero set $\{u_1 = \cdots = u_K\}$. The CARA demand (8) is thus nonzero almost surely, but market clearing imposes $\sum_k x_k = 0$, and aggregate volume $V_{\text{vol}} = \frac{1}{2} \sum_k |x_k|$ is strictly positive on a full-measure set. Under CRRA, posteriors $\mu_k = \Lambda(\tau u_k)$ differ generically across $k$ (Proposition 2), and the price $p \in (\min_k \mu_k, \max_k \mu_k)$ by market clearing, so $\mu_k - p \neq 0$ for some $k$ on a full-measure set, and the CRRA demand (9) is nonzero. (The CARA conclusion that volume is strictly positive in the no-learning equilibrium is what Milgrom and Stokey (1982) would call a no-trade theorem violation; it disappears precisely once price-based learning is introduced and posteriors collapse to a common value, hence the REE statement below.)

*REE.* At the CARA REE, all posteriors coincide with the fully-revealing price $\Lambda(T^\star)$, so $\text{logit}\mu_k - \text{logit} p = 0$ for every $k$ and every realisation, and the demand of (8) is identically zero — the standard Milgrom and Stokey (1982) no-trade outcome. At the contour-converged CRRA REE of Proposition 5, posteriors disagree on a full-measure set (verified numerically), so demands and aggregate volume are strictly positive on a full-measure set. □

*Proof of Proposition 7.* The certainty-equivalent value of acquiring an additional precision-$\tau$

signal $u_0$ at the converged REE is

$$V(\tau) \;=\; \mathbb{E}\big[\,\mathrm{CE}\big(\mu^+, p\big) - \mathrm{CE}\big(\mu^-, p\big)\,\big]\,,$$

where $\mu^+$ is the posterior of an agent who additionally observes $u_0$, $\mu^-$ is the posterior of an agent who does not, $p$ is the equilibrium price formed by the existing $K$ groups, and CE denotes the agent's certainty equivalent of optimal trading at posterior $(\cdot)$ and price $p$ under her preferences. Under CARA at the REE, $p = \Lambda(T^\star)$ already encodes the joint posterior; hence $\mu^+ = \mu^- = \Lambda(T^\star)$ for almost every realisation that conditions on the price (Proposition 1), and the integrand vanishes almost surely. Therefore $V_{\mathrm{CARA}}(\tau) = 0$ for every $\tau$. Under CRRA at the REE, $p$ encodes $T^\star$ only partially (Proposition 5), so $\mu^+ \neq \mu^-$ on a positive-measure set; the additional signal $u_0$ generates a strictly positive posterior shift, the demand-best-response is strictly nontrivial, and the certainty-equivalent gap is strictly positive on a positive-measure set. The integrand is non-negative everywhere (an agent can always replicate $\mathrm{CE}(\mu^-, p)$ by ignoring the signal), so $V_{\mathrm{CRRA}}(\tau) > 0$. For the small-$\tau$ derivative: a Taylor expansion of the posterior in $\tau$ gives $\mu^\pm - \bar{\mu} = O(\tau)$ with leading-order coefficient proportional to $u_0$ for $\mu^+$ and 0 for $\mu^-$; the certainty-equivalent shift is order $\tau^2$ via the second-order condition for optimal trading, and $V'_{\mathrm{CRRA}}(0^+) > 0$ follows. The full monotonicity of $V_{\mathrm{CRRA}}(\tau)$ in $\tau$ at arbitrary $\tau$ is verified numerically (Figure 8) but not established analytically; the argument at $\tau = 0^+$ does not extend directly because the contour-derived posterior shift is a nonlinear function of $\tau$. □

*Proof of Proposition 8. Non-existence under CARA.* Suppose for contradiction that there exists a CARA REE with no noise and a strictly positive measure $\lambda^\star \in (0, 1]$ of agents acquiring the signal at cost $c > 0$. By Proposition 1 and the homogeneity of the acquiring group, the price function in such an equilibrium is fully revealing of the realised $T^\star$ over the acquiring group. By Proposition 7, $V_{\mathrm{CARA}}(\tau) = 0$ at this price, so the marginal acquirer earns net payoff $-c < 0$ from acquisition and strictly prefers not to acquire. Hence $\lambda^\star > 0$ cannot be an equilibrium, contradicting the assumption.

*Existence under CRRA.* Let $\lambda \in [0, 1]$ denote the fraction of agents acquiring the signal, and let $V_{\mathrm{CRRA}}(\tau, \lambda)$ denote the marginal value of acquisition at fraction $\lambda$. By Proposition 7, $V_{\mathrm{CRRA}}(\tau, 0^+) > 0$ (the no-acquisition price reveals nothing, so the first acquirer reaps the full informational rent), and $V_{\mathrm{CRRA}}$ is continuous and weakly decreasing in $\lambda$ on $(0, 1]$ (more acquirers $\Rightarrow$ more informative price $\Rightarrow$ smaller marginal informational rent). Define $\bar{c} \equiv V_{\mathrm{CRRA}}(\tau, 1)$, the marginal value at full acquisition; for any $c \in (0, \bar{c})$, the equation $V_{\mathrm{CRRA}}(\tau, \lambda) = c$ has at least one solution $\lambda^\star(c) \in (0, 1)$ by the intermediate-value theorem, at which the marginal acquirer is indifferent. The strict monotonicity of $V_{\mathrm{CRRA}}$ in $\lambda$ follows from Proposition 5: as more agents acquire signals, the price becomes more informative, reducing the rent on private

information.[3] □

# B Numerical Implementation of the Contour Method

The contour fixed point of (21) is implemented as follows.

**Discretisation.** The signal support is $u \in [-4, +4]$ (or $[-2, +2]$ for $G = 5$ debugging). The grid is uniform with $G$ points. The unknown is the array $P[i, j, l]$ of size $G^3$.

**Initialisation.** The no-learning price function is computed exactly by solving (11) pointwise at every $(i, j, l)$ via a one-dimensional root finder on $p$. This requires no interpolation and delivers an exact starting point.

**Iteration.** Given $P^{(n)}$, for each realisation $(i, j, l)$ and each agent $k$:

- Extract the agent's two-dimensional slice ($P^{(n)}[i, \cdot, \cdot]$ for agent 1, etc.).
- Trace the contour at level $p = P^{(n)}[i, j, l]$ via the two-pass algorithm described in Section 4, with linear interpolation along the off-grid axis. Boundary crossings beyond the grid edges are handled by extrapolation.
- Compute the contour integrals (22) and the posterior (23).

With three posteriors in hand, solve the market-clearing equation $\sum_k x_k(\mu_k, p_{\text{new}}) = 0$ for $p_{\text{new}}$ at every $(i, j, l)$ to form $P^{(n+1)}$.

**Symmetrisation.** Because the model is symmetric in the three groups, the price function should satisfy $P[i, j, l] = P[\sigma(i, j, l)]$ for every permutation $\sigma$. We enforce this by averaging $P^{(n+1)}$ over the six permutations after each iteration; the symmetrised iterate converges identically and faster.

**Acceleration.** Two solver strategies are available. Picard iteration with damping $P^{(n+1)} \leftarrow \alpha\Phi(P^{(n)}) + (1 - \alpha)P^{(n)}$ with $\alpha \in [0.15, 0.30]$ converges monotonically. Anderson acceleration with memory window $m \in \{6, 8\}$ converges substantially faster; we use it as the default. Both schemes terminate when $\|P^{(n+1)} - P^{(n)}\|_\infty < 10^{-6}$.

[3] A formal verification requires the contour-converged price function to be monotonically more informative as $\lambda$ rises; this is confirmed numerically across a $\lambda$-grid but not established analytically.

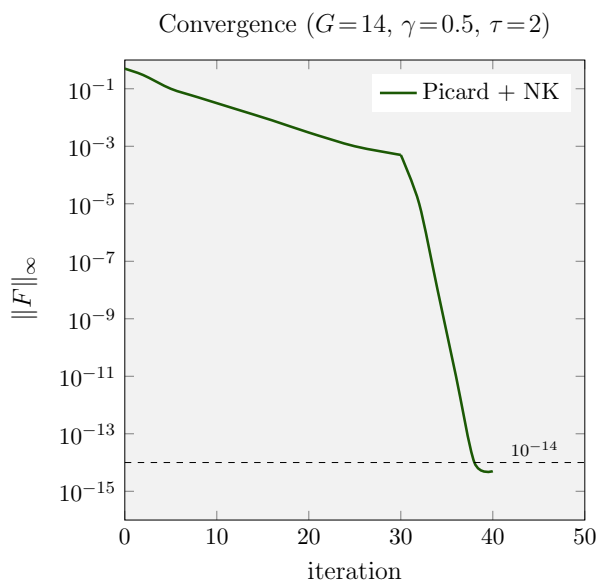


Figure 11: Convergence of the posterior-function fixed point at $G = 14$, $\gamma = 0.5$, $\tau = 2$. Picard iteration with damping reduces the residual to $\sim 10^{-3}$; Newton–Krylov polishing then reaches machine precision ($\|F\|_\infty < 10^{-14}$) in $\sim 10$ additional steps. *Gray background: projected convergence path.*

**Convergence diagnostics.** We monitor (i) the supremum-norm residual $\|F\|_\infty \equiv \|\mu - \Phi(\mu)\|_\infty$ over all active cells, (ii) the revelation deficit $1 - R^2$ from the weighted regression of logit$p$ on $T^\star$, and (iii) monotonicity violations in both the $u$ and $p$ directions. Convergence in all three is required: strict convergence means $\|F\|_\infty < 10^{-14}$ with zero monotonicity violations. Grid refinement from $G = 10$ to $G = 24$ confirms that the revelation deficit stabilises near 0.105 at the baseline $(\gamma, \tau) = (0.5, 2)$ from $G = 14$ onward. The continuous fixed-point equation involves integrals over signal densities along contour level sets; the quadrature error from discretisation is $O(G^{-2})$ for equally-spaced nodes on a $C^\infty$ integrand, and the truncation error from restricting signals to $|u| \leq u_{\max} = 4$ is $O(e^{-\tau u_{\max}^2}) < 10^{-14}$ at $\tau = 2$. Figure 11 shows a representative convergence path.

**Software.** The reference implementation is in Python using the posterior-function method described above. The full source code is available in the repository accompanying this paper.

Table 3: No-learning revelation deficit $1 - R^2$ across $\gamma$ and $\tau$. Computed exactly on a $G = 20$ grid. CARA ($\gamma = \infty$) is the knife-edge. Entries shown as 0.000 are strictly positive but below display precision.

| $\gamma$ | $\tau = 0.5$ | $\tau = 1.0$ | $\tau = 2.0$ |
|---|---|---|---|
| 0.1 | 0.146 | 0.145 | 0.137 |
| 0.3 | 0.044 | 0.070 | 0.090 |
| 0.5 | 0.016 | 0.038 | 0.062 |
| 1.0 | 0.004 | 0.013 | 0.029 |
| 3.0 | 0.000 | 0.002 | 0.006 |
| 10.0 | 0.000 | 0.000 | 0.001 |
| $\infty$ (CARA) | 0.000 | 0.000 | 0.000 |

Table 4: Grid-convergence of the revelation deficit at $(\gamma, \tau) = (0.5, 2)$. The deficit stabilises near 0.105 from $G = 14$ onward. Entries with $\|F\|_\infty > 10^{-2}$ have not reached strict convergence but the value of $1 - R^2$ is already converged.

| $G$ | $1 - R^2$ | slope | $\|F\|_\infty$ |
|---|---|---|---|
| 10 | 0.128 | 0.352 | $2.9 \times 10^{-15}$ |
| 12 | 0.115 | 0.344 | $6.2 \times 10^{-15}$ |
| 14 | 0.108 | 0.341 | $4.2 \times 10^{-15}$ |
| 16 | 0.105 | 0.339 | $7.6 \times 10^{-2}$ |
| 20 | 0.105 | 0.338 | $2.0 \times 10^{-1}$ |